復健相關解剖學及生理學

國立臺灣大學附設動物醫院

復健及整合醫學科

主治獸醫師

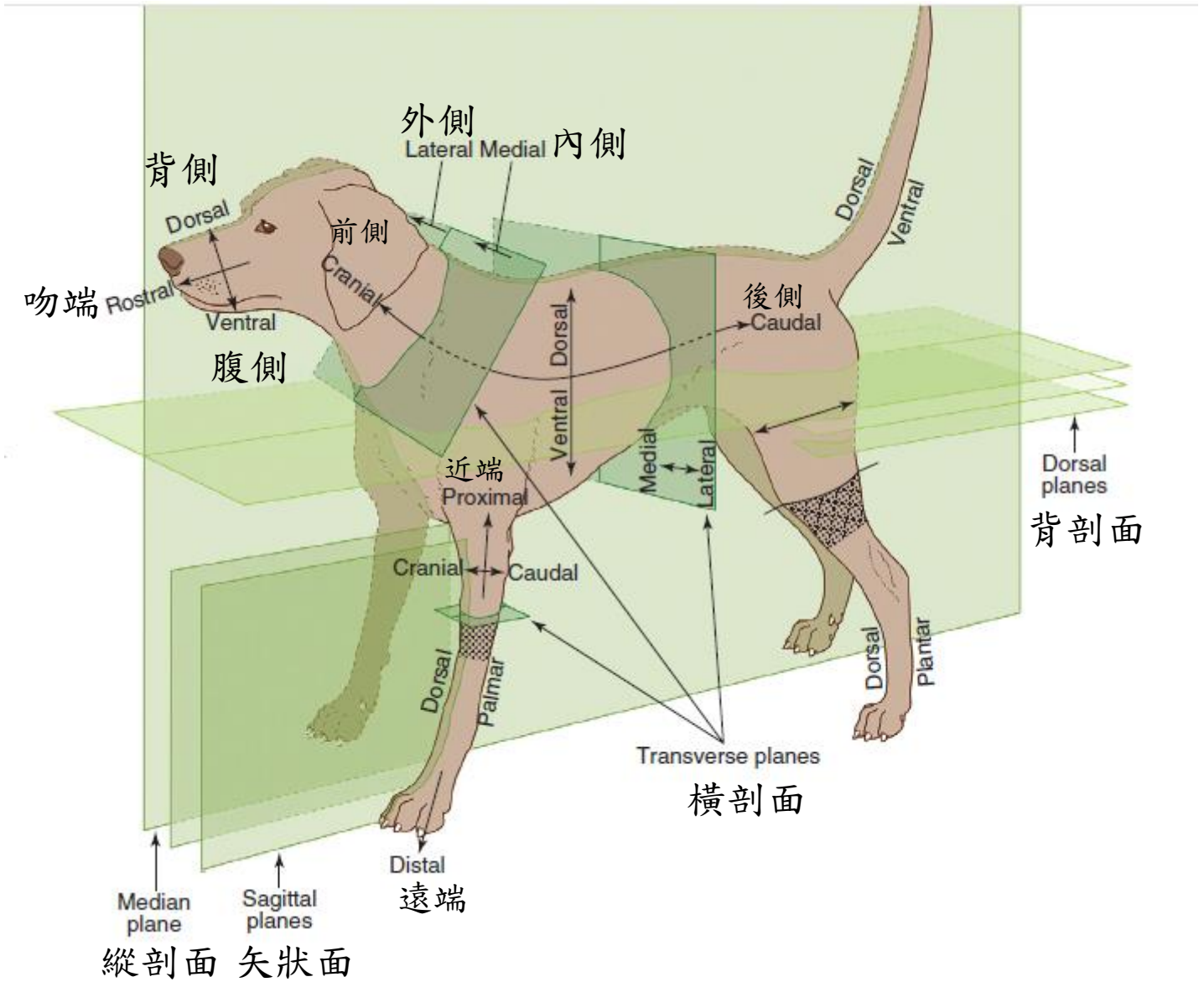
劉金鳴

CVA, CVCH, CVFT, CVTP,

CCRP, CKTP, CVPP

DACVSMR (Canine) Candidate

©Liu, Ching-Ming 2026



©Liu, Ching-Ming 2026

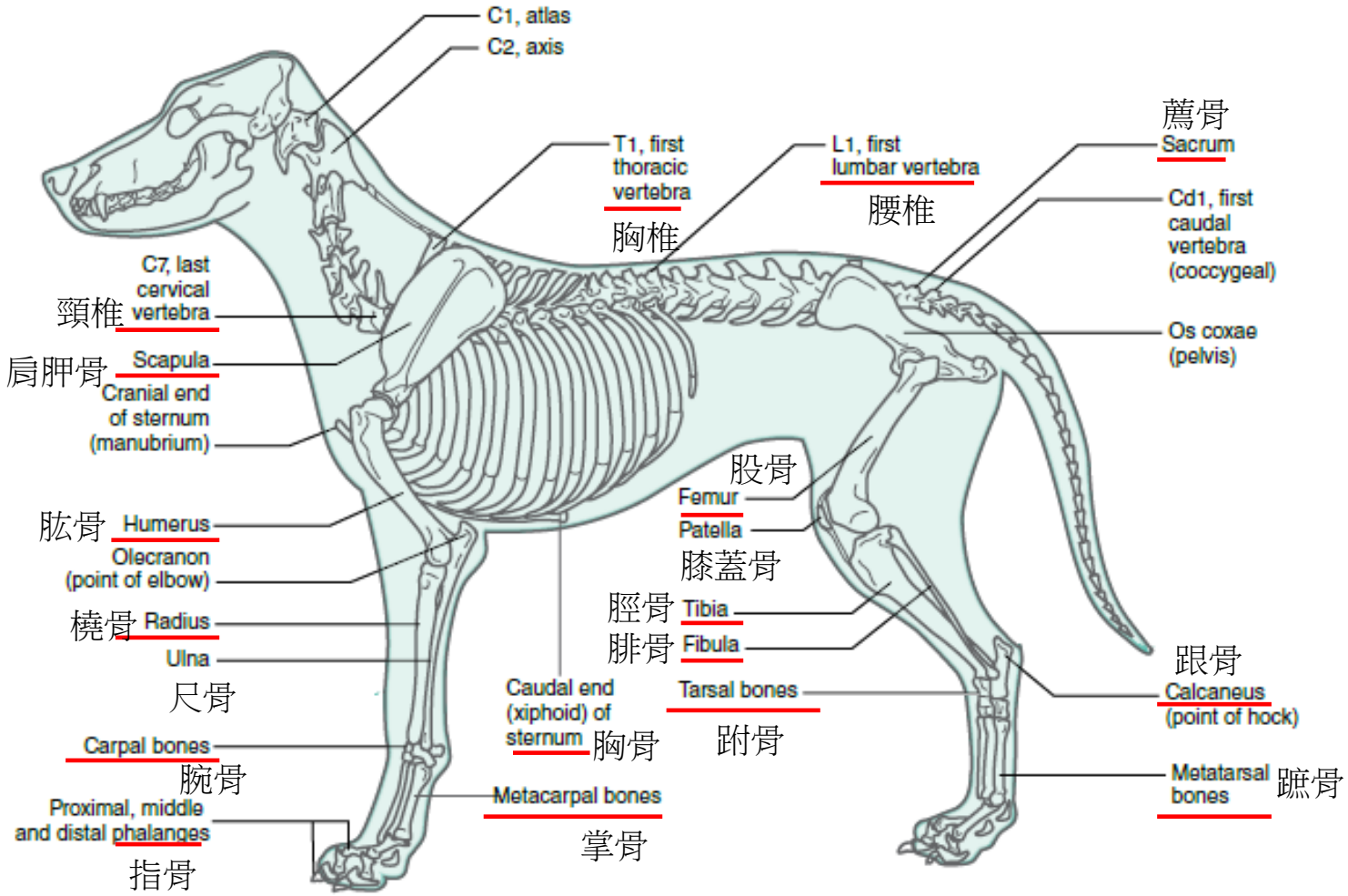
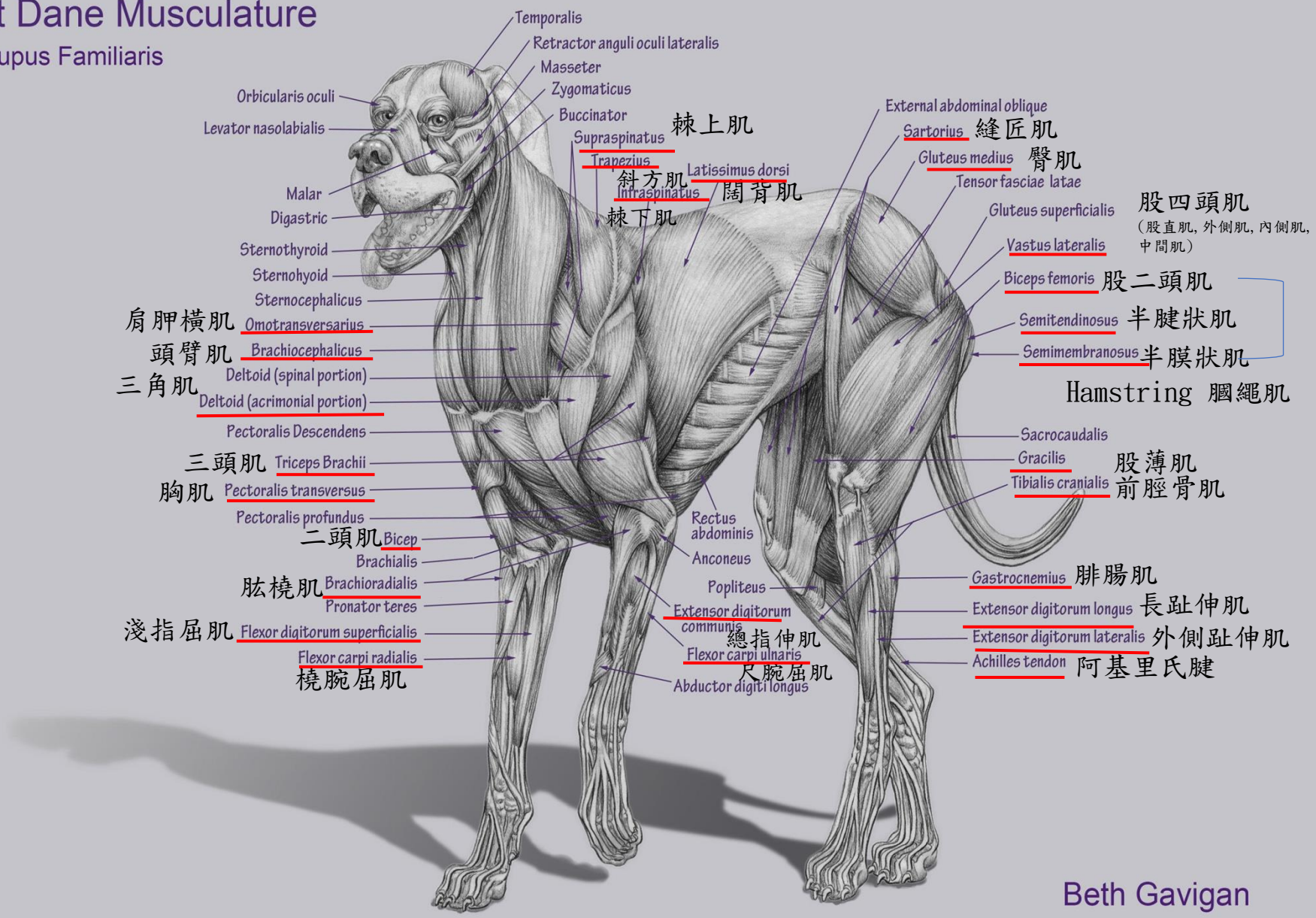


Fig. 3.1. The skeleton of the dog showing the major bones.
(With permission from T Colville, JM Bassett, 2001. Clinical anatomy and physiology for veterinary technicians. St Louis, MO: Mosby, p. 102.)

©Liu, Ching-Ming 2026

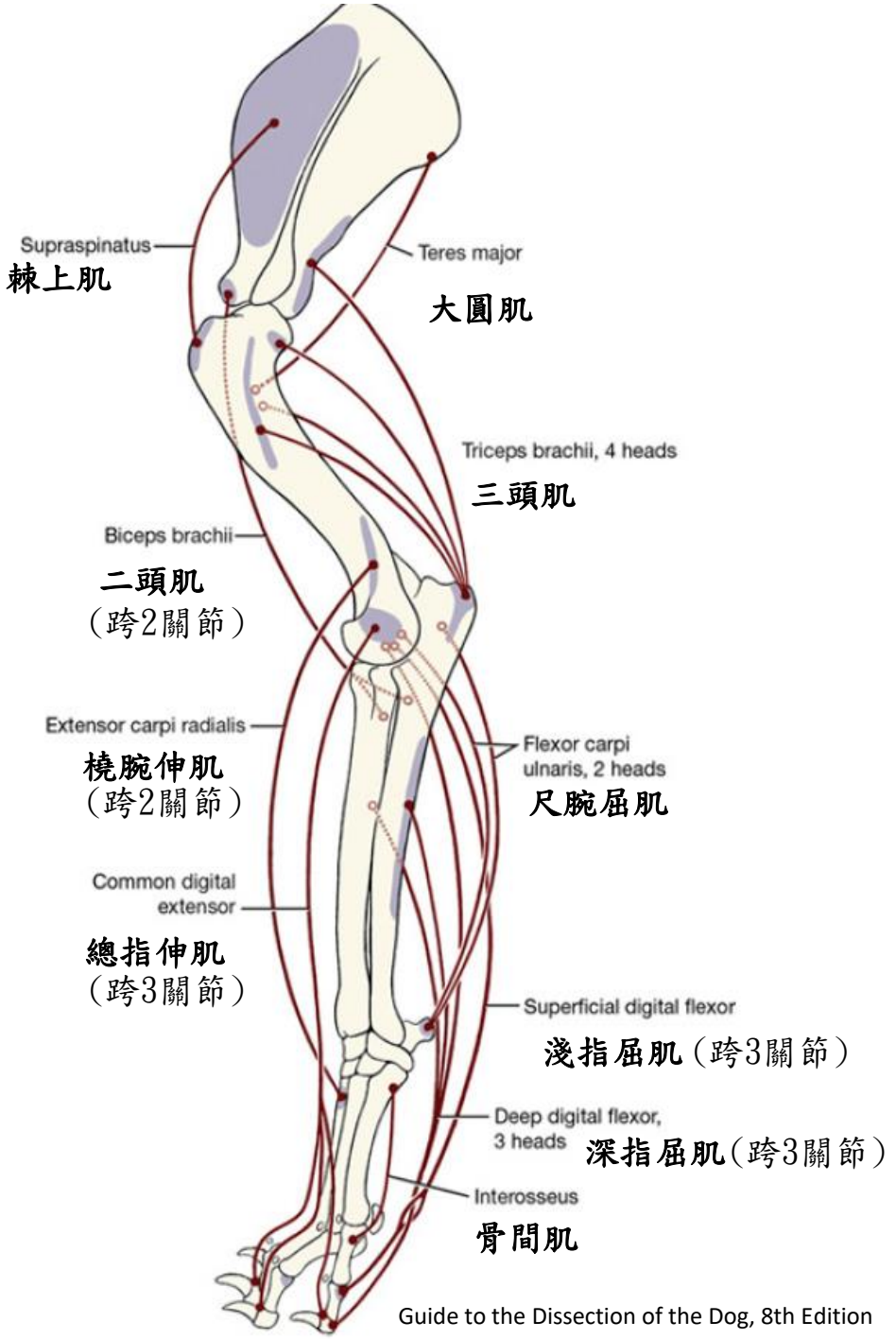
Great Dane Musculature

Canis Lupus Familiaris



©Liu, Ching-Ming 2026

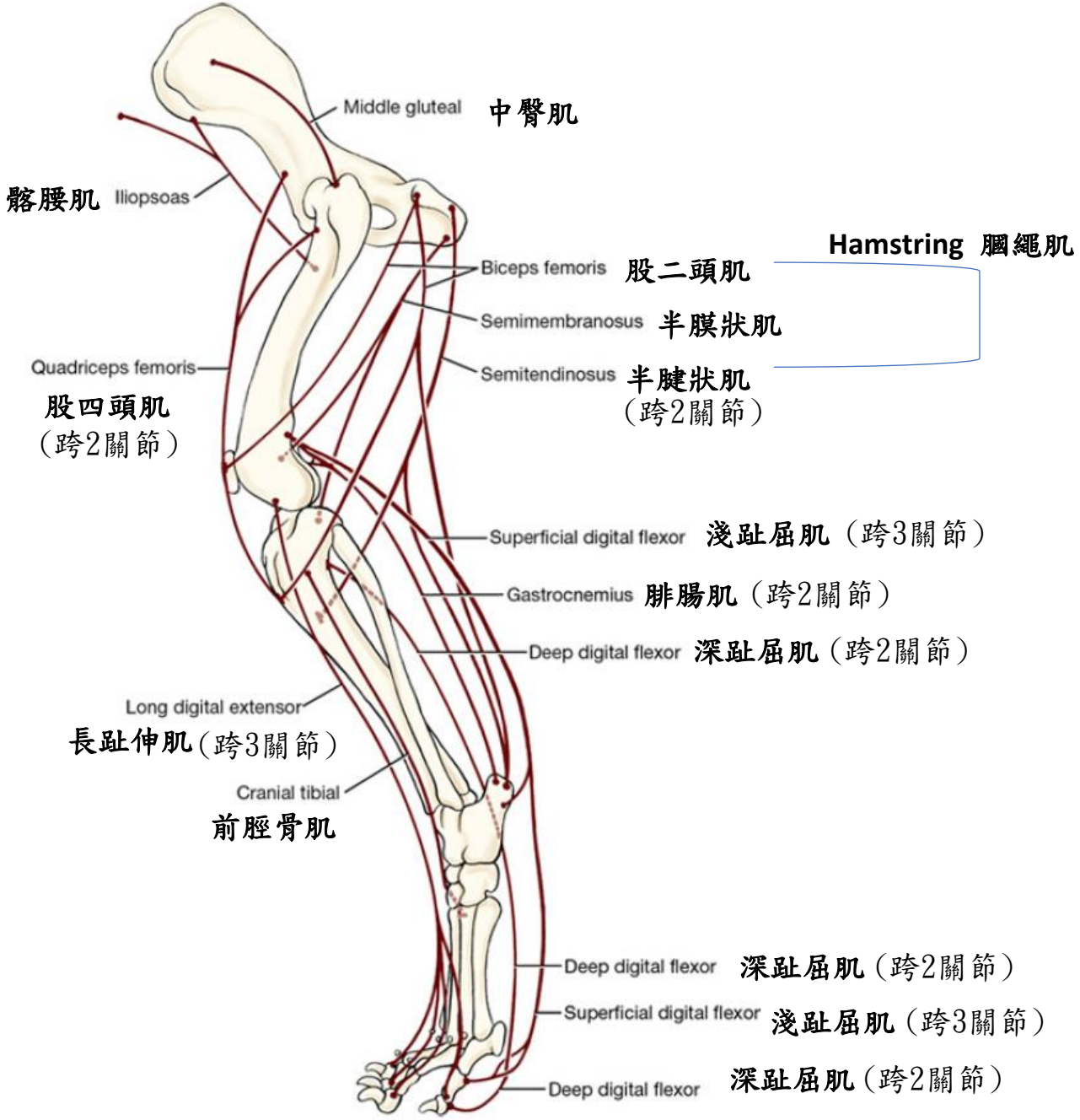
前肢



Guide to the Dissection of the Dog, 8th Edition

©Liu, Ching-Ming 2026

後肢



Guide to the Dissection of the Dog, 8th Edition

©Liu, Ching-Ming 2026

關節伸屈方向

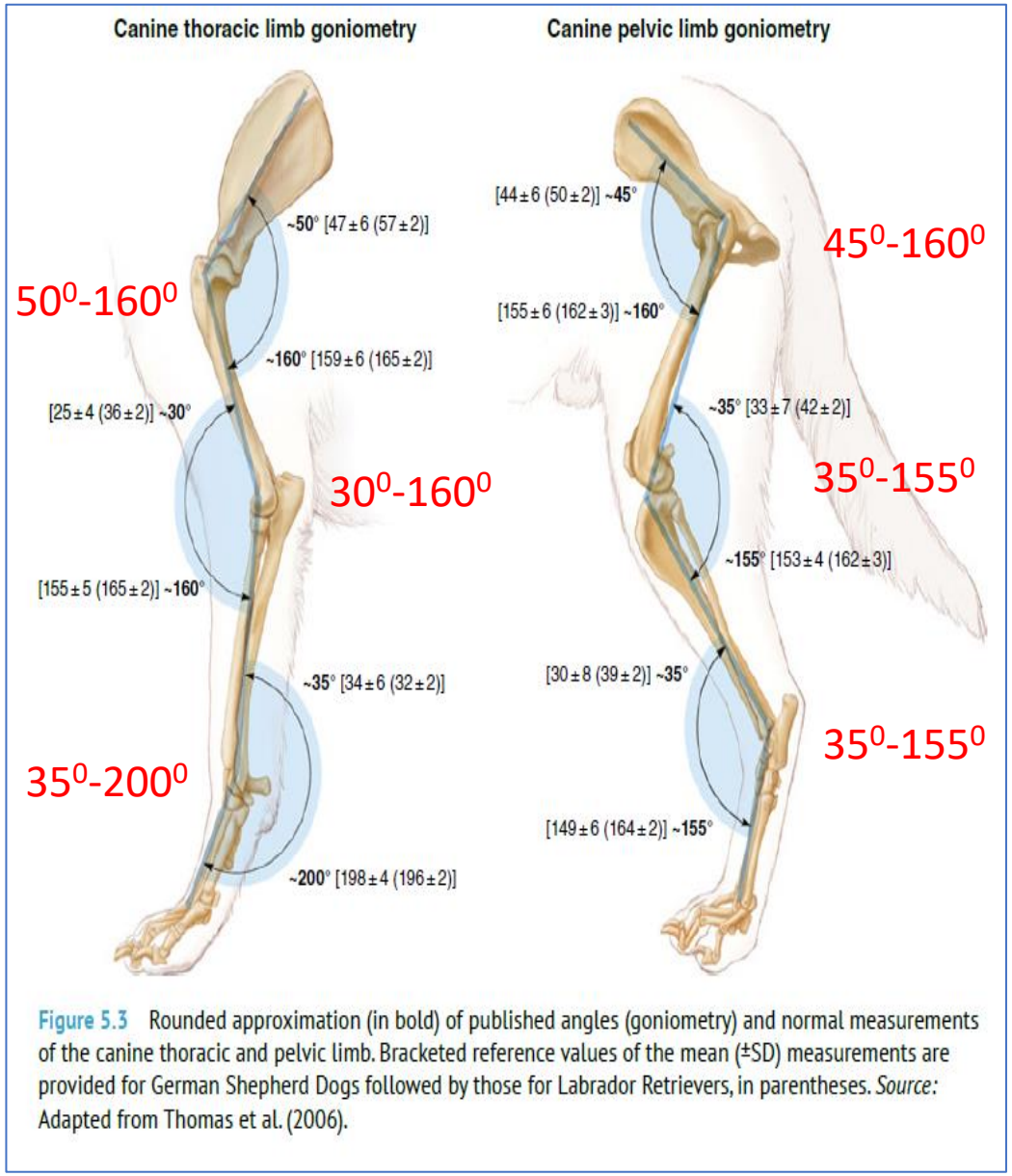


Figure 5.3 Rounded approximation (in bold) of published angles (goniometry) and normal measurements of the canine thoracic and pelvic limb. Bracketed reference values of the mean (±SD) measurements are provided for German Shepherd Dogs followed by those for Labrador Retrievers, in parentheses. *Source:* Adapted from Thomas et al. (2006).

Duerr 2020, Canine Lameness

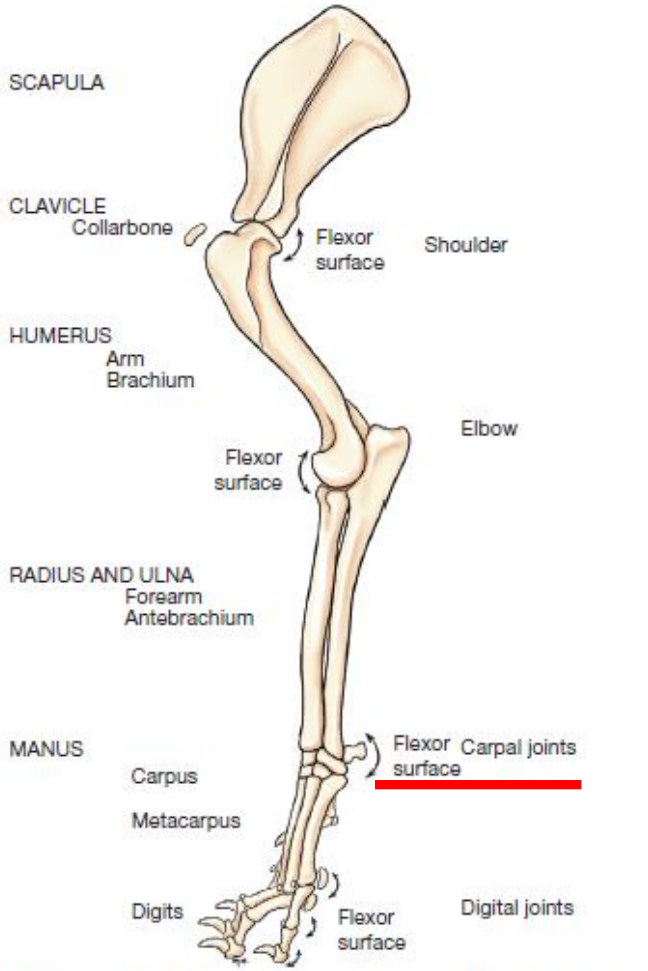


Figure 5-3 Left forelimb skeleton, noting joints and flexor surfaces. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog*, ed 7, Philadelphia, 2010, WB Saunders.)

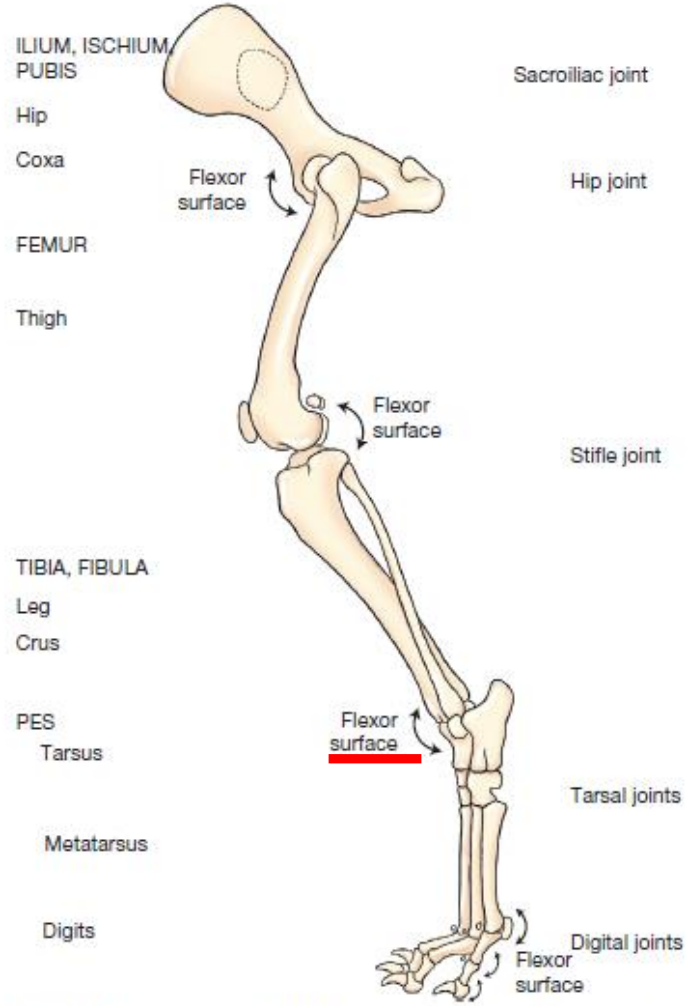


Figure 5-4 Left hindlimb skeleton, noting joints and flexor surfaces. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog*, ed 7, Philadelphia, 2010, WB Saunders.)

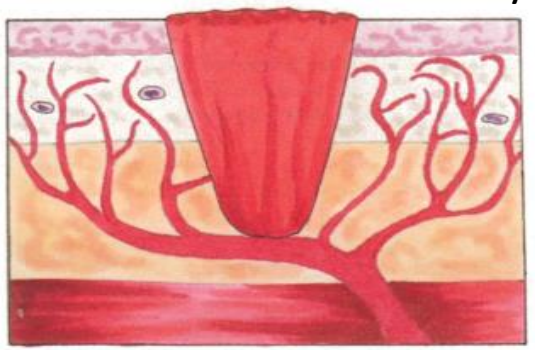
©Liu, Ching-Ming 2026

Wound healing 傷口癒合

- **Regeneration 再生**
  - Restore the original properties and function e.g. bone healing
- **Repair 修補**
  - Scar tissue as replacement e.g. cartilage, muscles, ligaments, tendons healing
- **Phases of wound healing**
  - Inflammatory**發炎期(day 0-5)**
    - Rehabilitation limited only modalities to speed up healing, reduce inflammation **僅能使用減少炎症加速癒合復健設備**
    - **Avoid active exercises 避免主動運動**
  - Reparative**修補期(day 5-21)**(proliferative, fibroblastic)
    - Exposure to physical stressors can improve the orientation and stability of the tissues**施加物理性壓力可增進組織排列及穩定度**
  - Remodeling**重建期(day 21~a year)** Consolidation and maturation
    - The patient may be subjected to an increased level of stress **能承受更多應力**

Inflammatory phase發炎期 2nd - 5th day

Vascular section 0-2nd day



cellular section 2nd -5th day



Reparative 修補期
phase 5th -21st day



Remodeling重建期
phase 21st day~



Essential facts of physical medicine, rehabilitation and sports medicine in companion animals, Bockstahler 2019

©Liu, Ching-Ming 2026

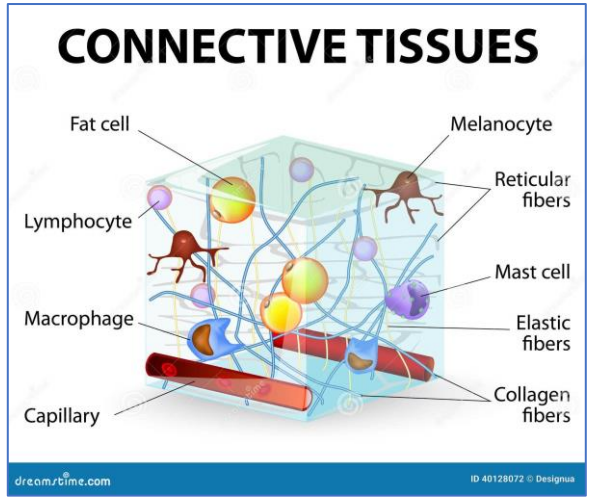
Connective tissue 結締組織

- Function
  - Connection of tissues(連接) e.g. joint capsules, tendons, ligaments
  - Support and protection(支持及保護) e.g. bones and skeleton
  - Nutrition and waste transportation(營養及廢物運送)
  - Defense of the body(身體防禦)

- Types
  - Fibrous connective tissue 纖維組織(e.g. skin, tendons肌腱, ligaments韌帶, aponeurosis腱膜)
  - Adipose connective tissue 脂肪組織
  - Skeletal tissue 骨架組織 (e.g. bone, cartilage)

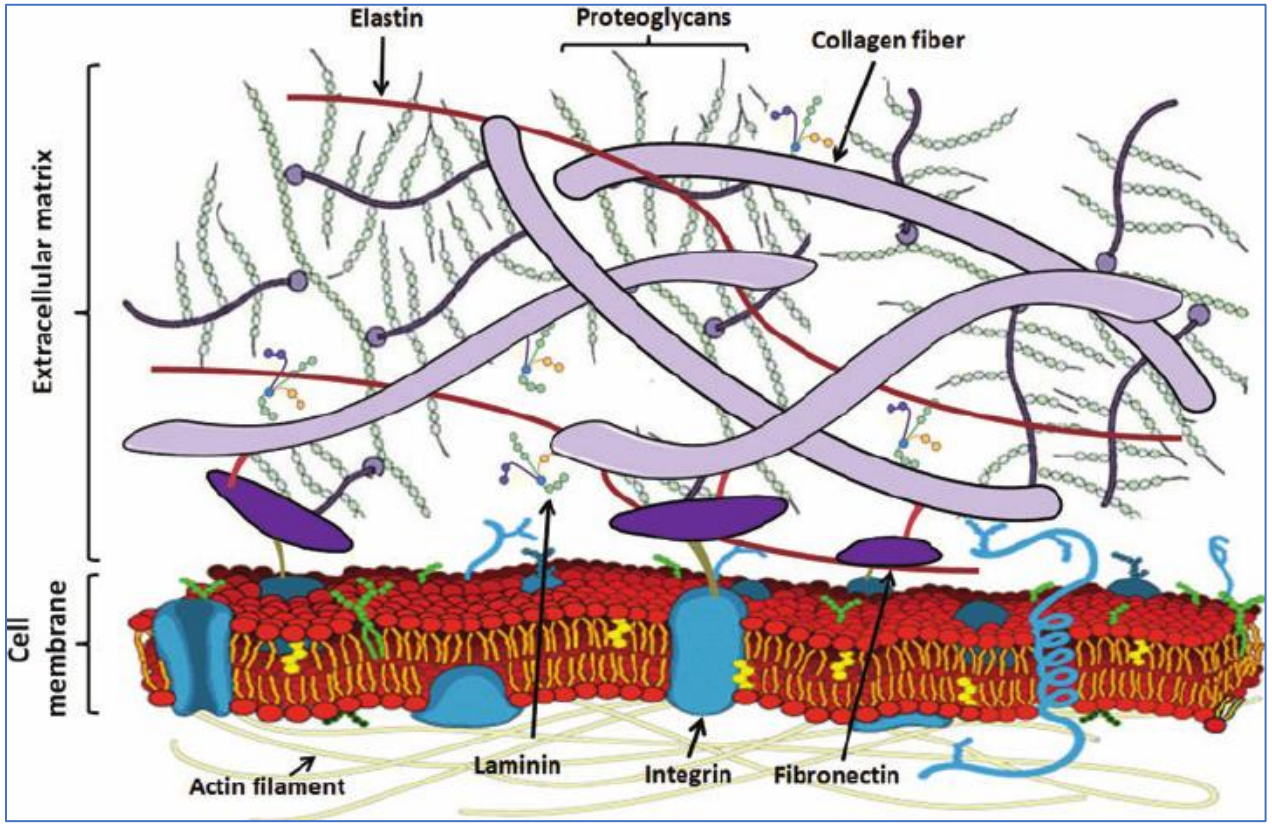
©Liu, Ching-Ming 2026

Connective tissue 結締組織



- **Cells**
  - Fibroblasts/fibrocytes **纖維母細胞/纖維細胞**
  - Osteoblasts/osteocytes **成骨細胞/骨細胞**
  - Chondroblasts/chondrocytes **軟骨母細胞/軟骨細胞**
  - Mast cells **肥大細胞**
  - Adipocytes **脂肪細胞**
  - Macrophages **巨噬細胞**
  - Leucocytes **白血球**

- **Extracellular matrix (ECM)細胞外基質**
  - Collagen fibers **膠原纖維**
  - Elastic fibers **彈性纖維**
  - Ground substance **基質**
  - Water **水**
  - Non-collagen fibers **非膠原纖維** e.g. cross-linking protein 交連蛋白



From google

©Liu, Ching-Ming 2026

- **Collagen fiber** 膠原纖維
  - **Type I**
    - Ligaments, tendons, capsules and bone
    - **Tensile stresses**
    - **負責承受伸展力**
  - **Type II**
    - Cartilage , intervertebral discs and menisci
    - **Pressure force**
    - **負責承受壓縮力**
  - **Type III**
    - Skin, subcutaneous tissue and synovial membrane
    - **Wound healing**
    - **負責傷口癒合**

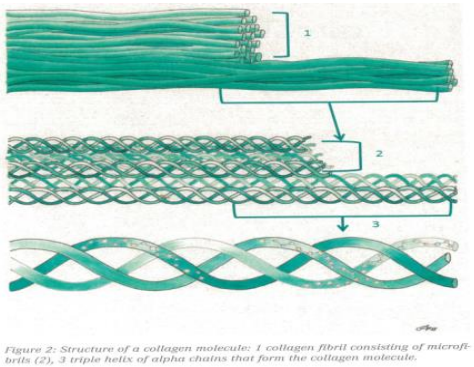


Figure 2: Structure of a collagen molecule: 1 collagen fibril consisting of microfibrils (2), 3 triple helix of alpha chains that form the collagen molecule.

Essential facts of physical medicine, rehabilitation and sports medicine in companion animals, Bockstahler 2019

◆ Gives Connective tissue stability and structure

穩定性及結構

◆ Absorbs tensile forces (e.g. ligaments and tendons)

吸收伸展力

◆ Absorbs compressive forces (e.g. cartilage and intervertebral disc)

吸收壓縮力

◆ Can be elongated by about 5%

可延長5%

◆ Fast and abrupt forces increase likelihood of injury. Take into consideration when performing physiotherapy exercises/stretching

物理治療及伸展時須注意勿急速施力因而造成損傷

◆ Increased stress causes collagen to become thicker and more stable

增加應力可使膠原纖維便後及更加穩定

◆ Immobilization diminishes mechanical loading capacity

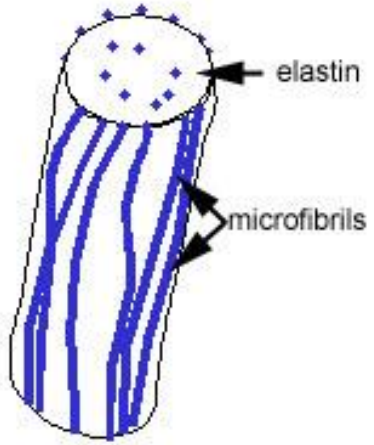
固定不動會減少承受負載能力

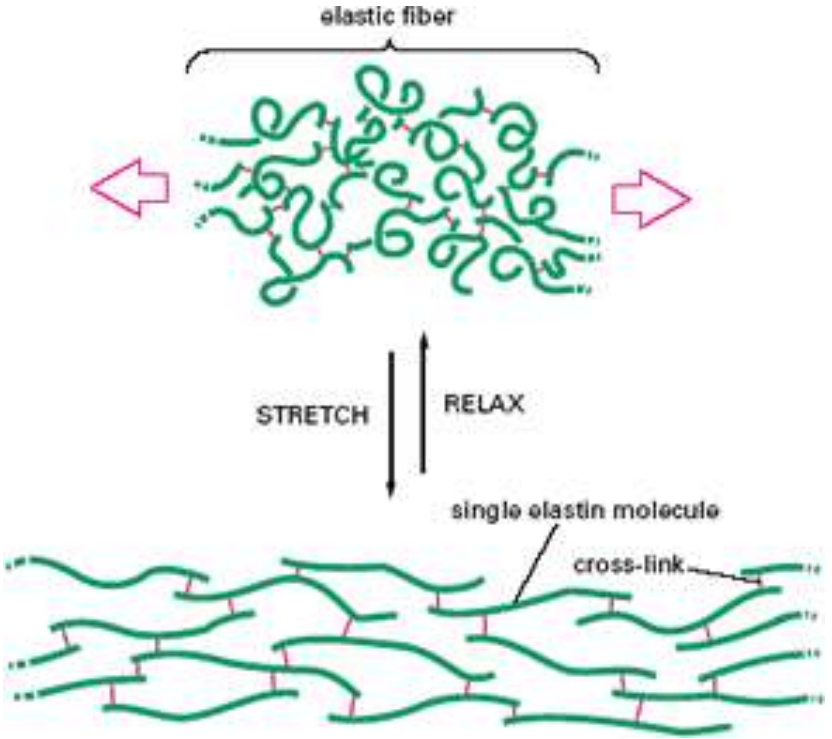
©Liu, Ching-Ming 2026

- **Elastic Fibers and elastin**

彈性纖維及彈性蛋白

- Elastic fibers are yellow color and found adjacent to the skin, the blood vessels and in loose connective tissue. Small percentages are also found in tendons and ligaments.
- **彈性纖維呈現黃色分佈於皮膚、血管、疏鬆結締組織，少部分於肌腱及韌帶組織**
- Elastic fibers are highly branched and form a meshwork-like structure. They are composed of elastin and are surrounded by microfibrils around elastin.
- **彈性纖維為網狀結構由微纖維圍繞彈性蛋白**
- The fibers are elastic and can be extended by 70% maximally up to 220%.
- **延伸力可由 70% 高達 220%**
- When tissue is subjected to force, the elastic fibers initially assume the load, then slowly transfer it to collagen fibers
- **組織承受力量時先由彈性纖維承受負載，然後將負載轉移給膠原纖維**

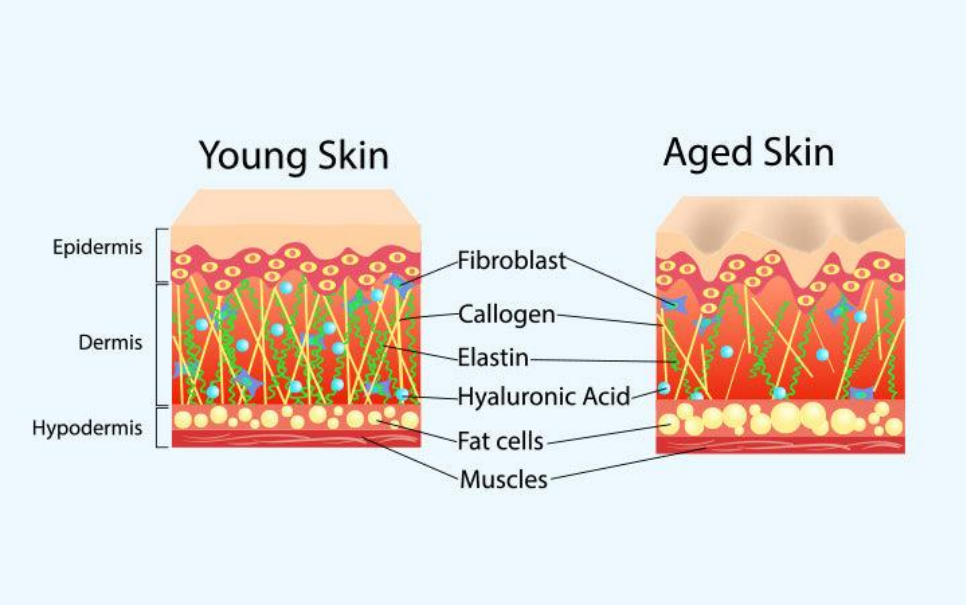


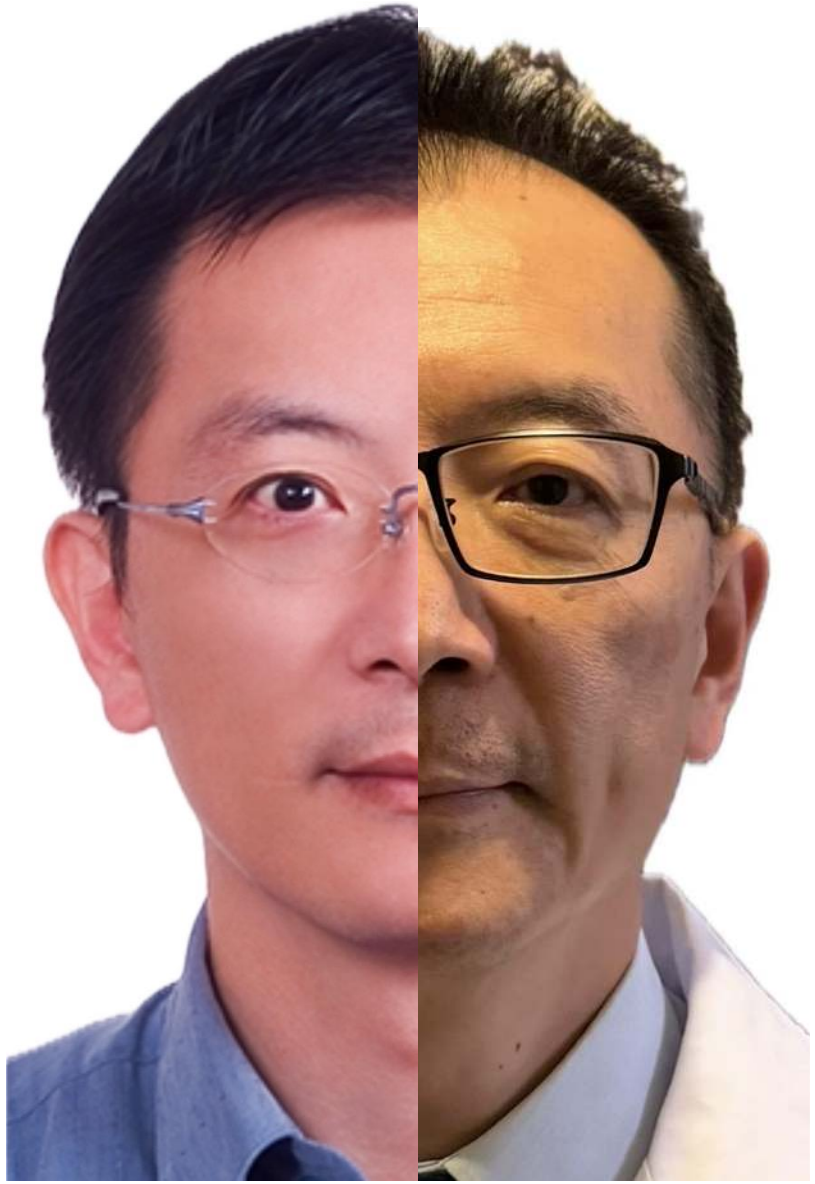


From google

©Liu, Ching-Ming 2026

- 彈性纖維與膠原纖維
  - 彈性纖維或稱為黃纖維能增強結締組織的彈性，並與膠原纖維共同提供組織的抗張能力。
  - 膠原纖維稱或為白纖維主要存在於韌帶、皮膚真皮層結締組織中，賦予皮膚彈性並提供強大的韌性與抗拉力，但缺乏彈力。

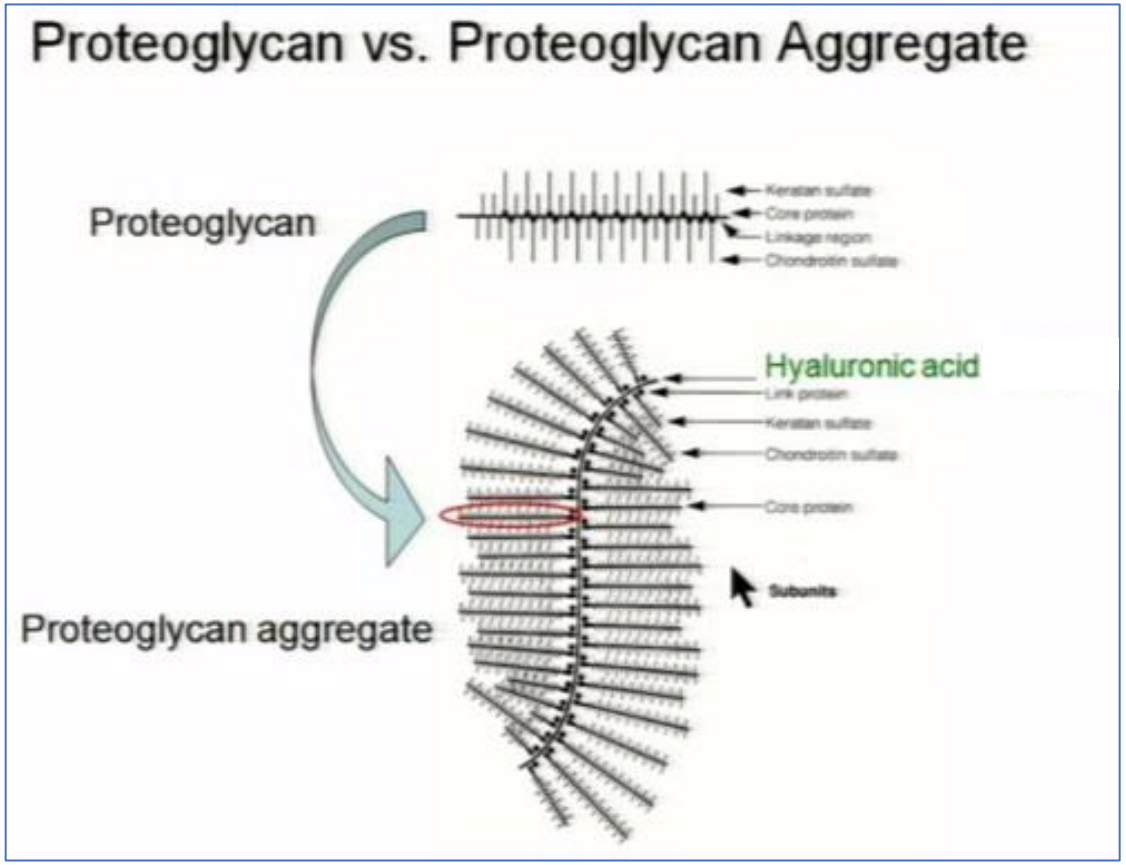




©Liu, Ching-Ming 2026

- **Ground substance 基質**

- Consists of proteoglycans aggregate (蛋白聚醣聚合體) having a very high affinity for water and bind to it in large quantities
- 含有蛋白聚醣聚合體，親水性強含水量高
- Absorb forces that act on tissue and protect it from excessive stress; most notable in cartilage, IVDs and menisci
- 吸收過多力量以保護組織例如軟骨、椎間盤、半月板
- GAG(醣胺聚醣) and PG (蛋白聚醣) combine with fibers to stabilize tissue
- 醣胺聚醣，蛋白聚醣結合纖維以穩定組織
- Contribute to elasticity of connective tissue allowing tissue to return to original shape after being subjected to force
- 提供組織彈性以因應外力回復原有形狀



From google

©Liu, Ching-Ming 2026

- **Nutrition營養**of connective tissue

- Oxygen, nutrition supply and waste product removal via transport system (capillaries, interstitial fluid) directly to cells
- **氧氣及營養供應及廢物排除透過微血管血液及組織液傳輸至細胞**

  - Blood circulation within tissues**組織內血液循環**
    - Perfused- Blood **血液**
    - non-perfused- Osmosis and diffusion **滲透及擴散**
  - Tissue loading and unloading **組織負重與不負重**
    - Mechanical stressor **機械應力**- circulation **血流循環**and piezoelectric **壓電效應**
      - Contracting and extending **收縮及伸展** (e.g. muscles and tendons)
    - Compressive forces **壓縮力** (e.g. cartilage, intervertebral disc and menisci)

©Liu, Ching-Ming 2026

Bone 骨

- Sections:
  - Articular cartilage 關節軟骨
  - Epiphysis 骨骺
  - Metaphysis 幹骺端
  - Diaphysis 骨幹
- Composition of bone 組成份
  - 5-10% water
  - 20% organic material (cellular and matrix)
  - **70% mineral ($Ca_5(PO4)_3(OH)$** – crystal of calcium and phosphorus)
- Structure
  - Compact bone 密質骨(Cortical bone)
  - Spongy bone 鬆質骨(Cancellous bone)
  - Harversian system 哈維氏系統
  - *Blood circulation* 血液循環
- The structure and design of bone has perfectly adapted to biomechanical stress
- 骨頭結構完全用以適應生物力學上之需求
- Bone is constantly undergoing remodeling and adapting to the demands placed on it.
- 骨頭隨需要不斷進行重塑以應付承重

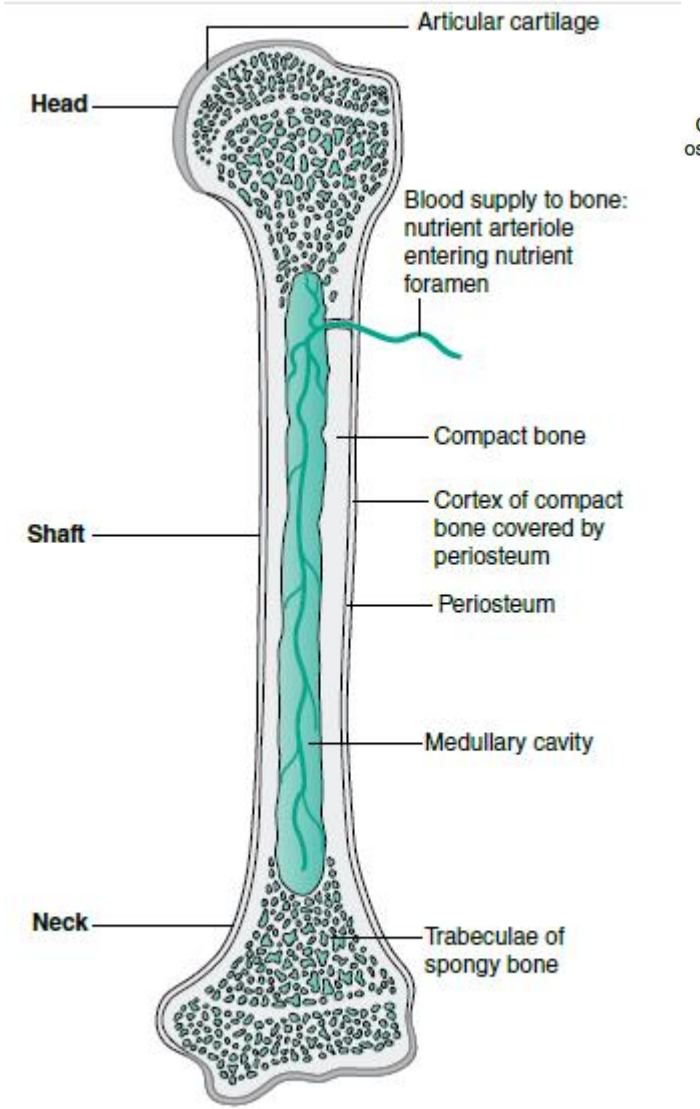
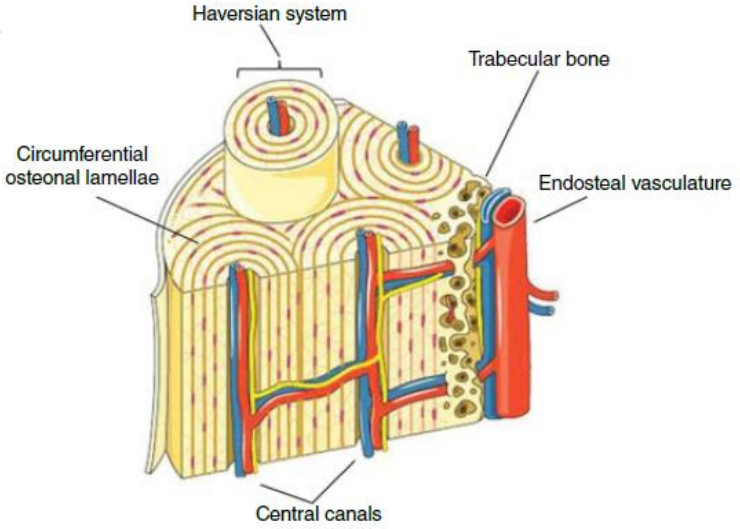
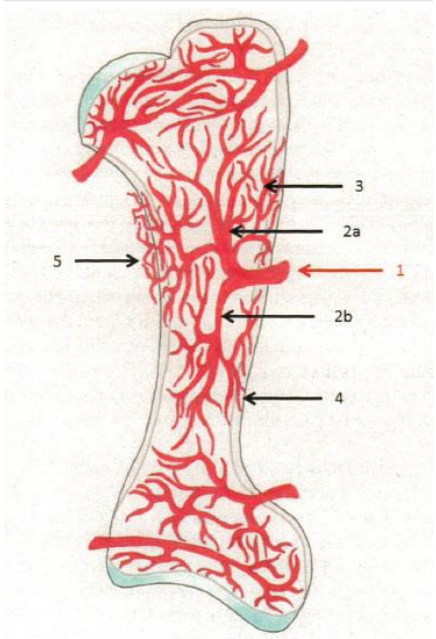


Fig. 3.2. Structure of a typical long bone.





Essential facts of physical medicine, rehabilitation and sports medicine in companion animals, Bockstahler 2019

©Liu, Ching-Ming 2026

Bone healing 骨癒合

- Direct (primary) bone healing 初級骨癒合
  - Gap length < 1mm, gap strain < 2%
    - Contact healing 直接接觸癒合
    - Gap healing: > 0.1 mm but < 1 mm
- Indirect (secondary) bone healing 次級骨癒合
  - Gap length > 1mm, gap strain > 2%
    - Callus formation following Classic tissue healing
    - 骨痂形成遵循組織癒合模式
    - **Wolff's law** 沃爾夫定律:

In a healthy individual bone will adapt to the forces under which it is placed in.

健康個體在特定骨骼的負載增加情形下，骨骼會慢慢的變強壯，來承受該負載

    - About 3 months remodeling
    - 約3個月重建期
- Bone remodeling- Piezoelectric effect 骨骼重建之壓電效應
  - +: on tensile force side the osteoclast increased
  - - : on compressive side the osteoblast increased
- Immobilization and remobilization
  - Slow leash walking or mild treadmill activity were recommended during the first month of remobilization
  - 在骨骼重建期第一個月，建議以牽繩行走或輕度跑步機活動

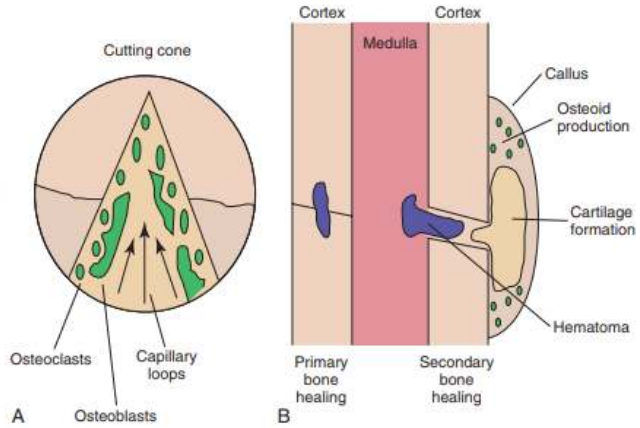
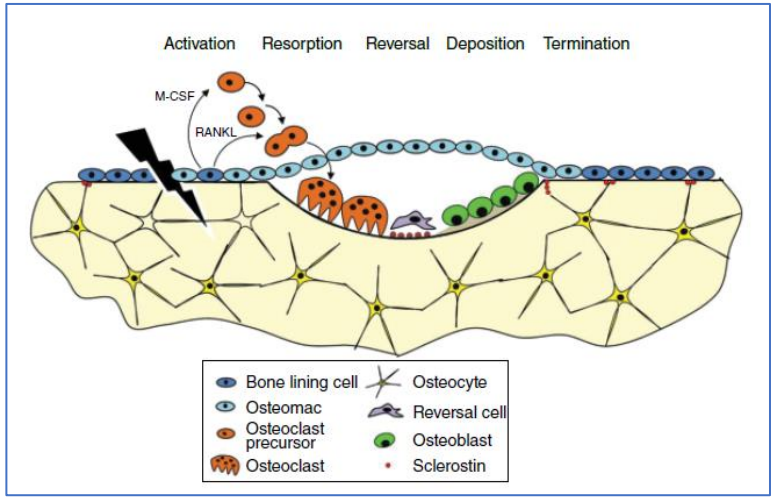
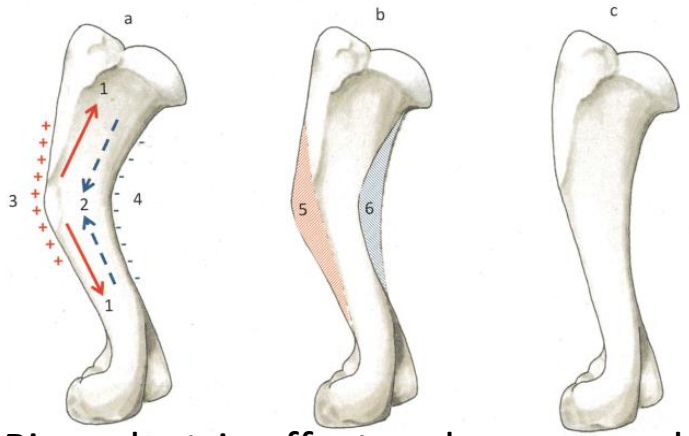


Figure 6-3 Bone healing. A, Primary bone healing with the formation of cutting cones. B, Secondary bone healing.





Piezoelectric effect on bone remodeling

©Liu, Ching-Ming 2026

- Factors Affecting Bone Healing 影響骨癒合的因素
  - Biomechanical factors 生物力學因子: Wolf's law
  - Electrical field 電場效應
  - Endocrine 內分泌: PTH, calcitonin, Vitamin D, thyroid
  - Age 年齡: young > senior
  - Therapeutic ultrasound, extracorporeal shock wave
  - 治療性超音波，體外震波

©Liu, Ching-Ming 2026

Healing Timeline 癒合時間軸: Bone

- Inflammation
  - Primary and secondary bone healing follow similar timelines.
  - Immediate to 72 hours (hematoma and inflammation, secondary bone healing)
  - **初級骨癒合與次級骨癒合有相似炎症期約72小時**
- Repair
  - Radiographic healing (secondary): appearance of callus at 10-12 days, disappearance of fracture line within 30 days.
  - Disappearance of fracture line radiographically as early as 30 days with secondary bone healing. Primary bone healing takes longer.
  - **骨痂屬於次級骨癒合出現在10-12天，影像學骨折線30天消失，初級骨癒合較次級骨癒合慢消失**
- Remodeling/Maturation
  - Woven bone development over mineralized callus. Complete remodeling of the callus 90 days after fracture repair for a simple fracture.
  - Primary bone healing (direct union, no callus formation) produces less early stability than secondary, and requires implants to be in place for a longer time than with secondary healing.

  - **骨痂形成編織骨到完全重建需90天**
  - **初級骨癒合無法提供早期穩定度，必須靠植入物提供穩定，時間比次級骨癒合更久**

©Liu, Ching-Ming 2026

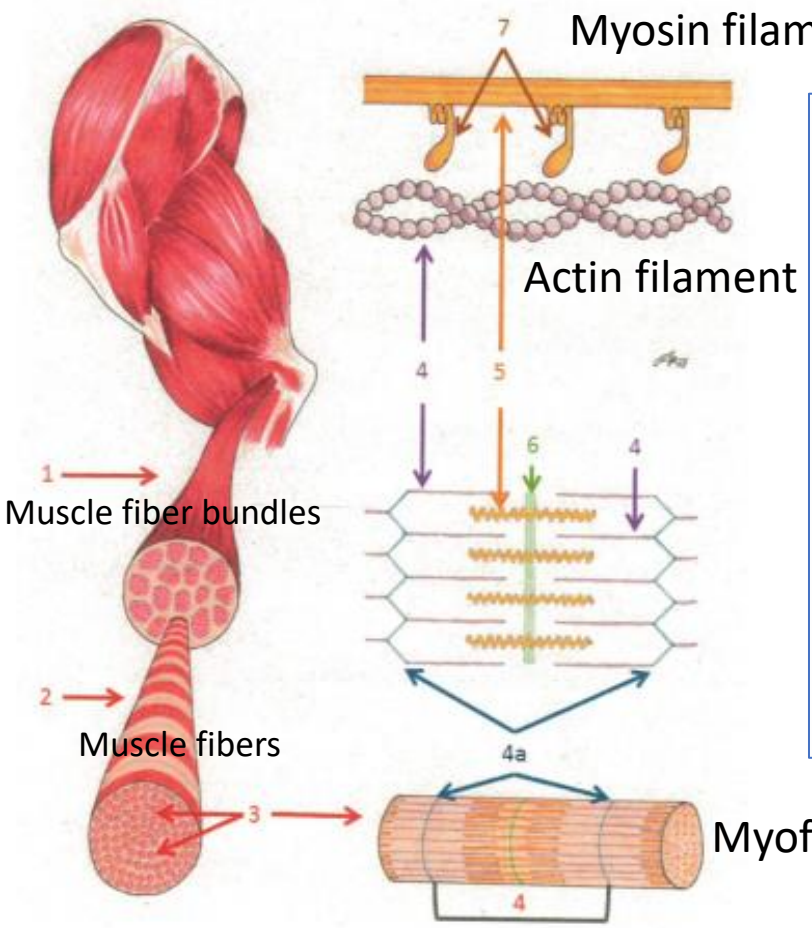
Implications for rehabilitation of bone骨骼復健的意涵

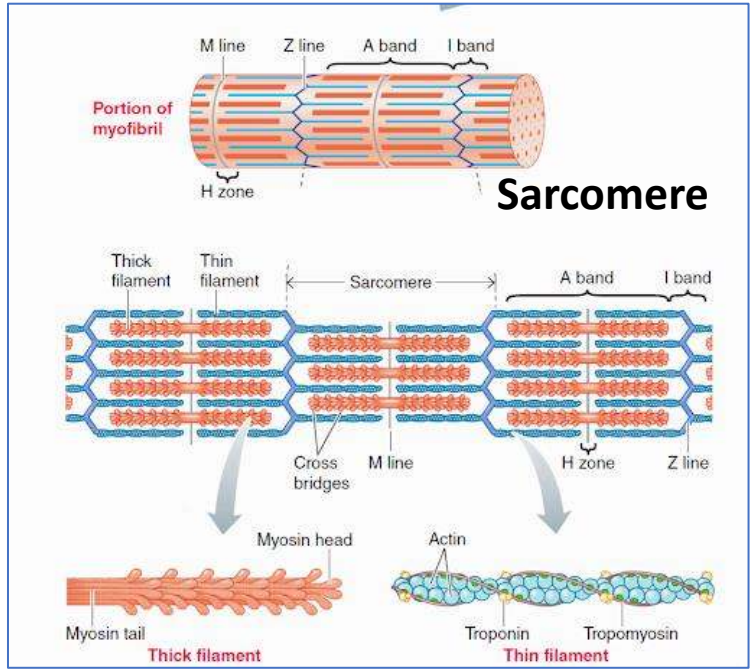
- During inflammatory phase (immediately up to 2-3 days after injury), decrease pain, minimize soft tissue inflammation, maintain joint health/Range of motion (ROM) and establish safe, controlled mobility, depending on fracture stability
- **在炎症期(3 天內)減少疼痛、減少發炎，骨折穩定前提下維持關節活動**
- During repair phase (3 days to 4-6 weeks after injury), small, gradually increasing cyclic loading to increase micromotion and improve callus formation with secondary healing; continued maintenance of joint health through ROM, active or passive
- **在修補期(3 天 to 4-6 週)逐漸增加骨負重增進骨痂形成進行次級骨癒合，持續關節活動維持關節健康**

©Liu, Ching-Ming 2026

Muscle 肌肉

- Muscle units
  - Muscle fiber bundles肌纖維束
  - Muscle fibers肌纖維
  - Myofibrils肌原纖維
  - Myosin filament肌球蛋白微絲
  - Actin filament肌動蛋白絲
  - Sarcomeres 肌小節
    - **Z** line
    - **M** line
    - **I** band
    - **A** band
    - **H** zone





Essential facts of physical medicine, rehabilitation and sports medicine in companion animals, Bockstahler 2019

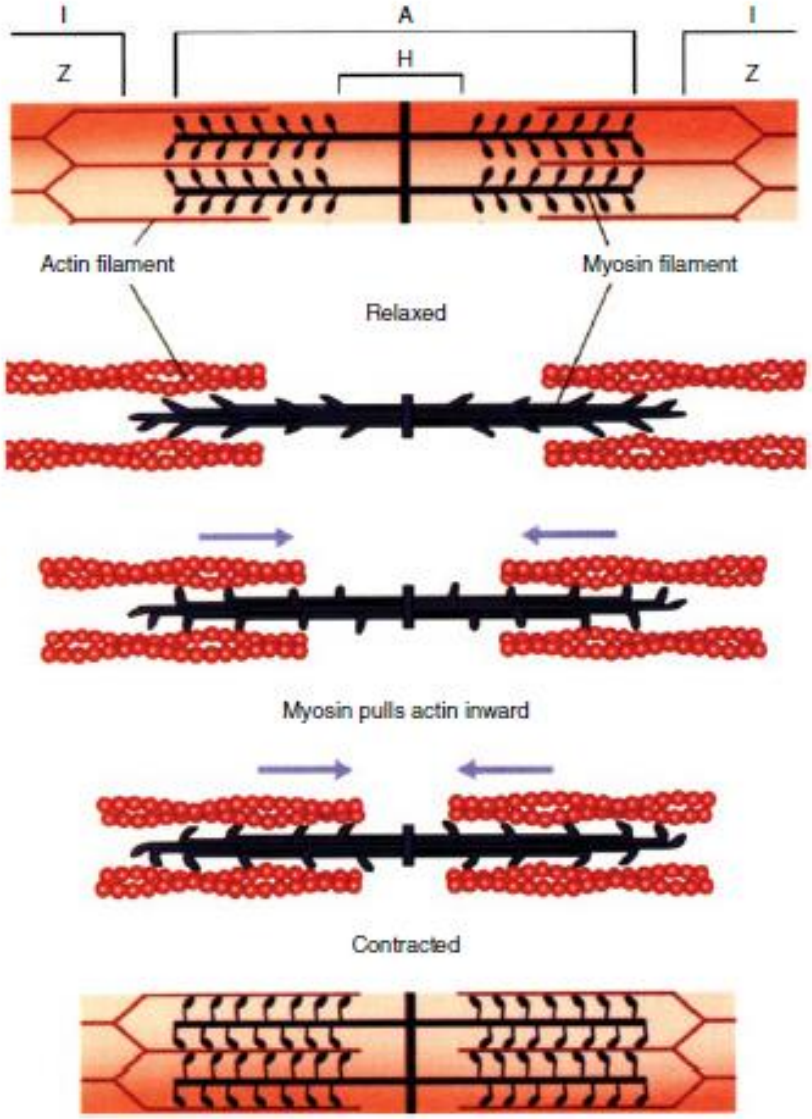


Figure 3.7 Contraction of skeletal muscle occurs due to ATP-dependent conformational changes of the globular heads of myosin proteins that cause myosin to slide along adjacent actin filaments. Source: Used with permission from Hand *et al.*, 2010.

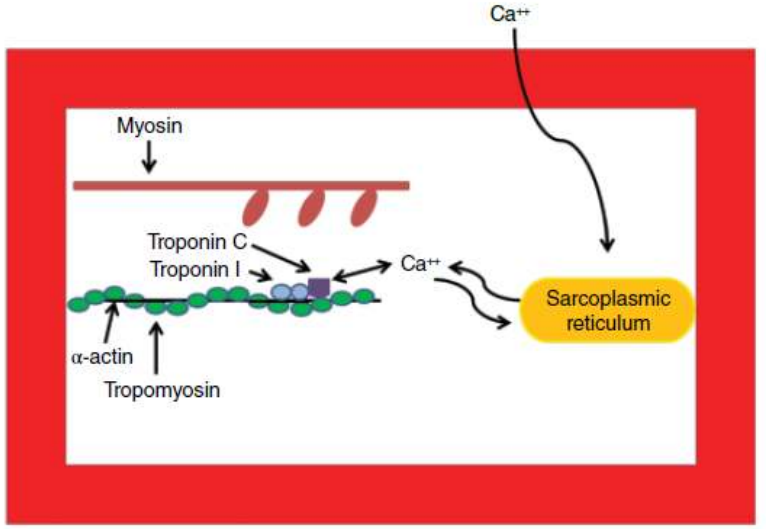
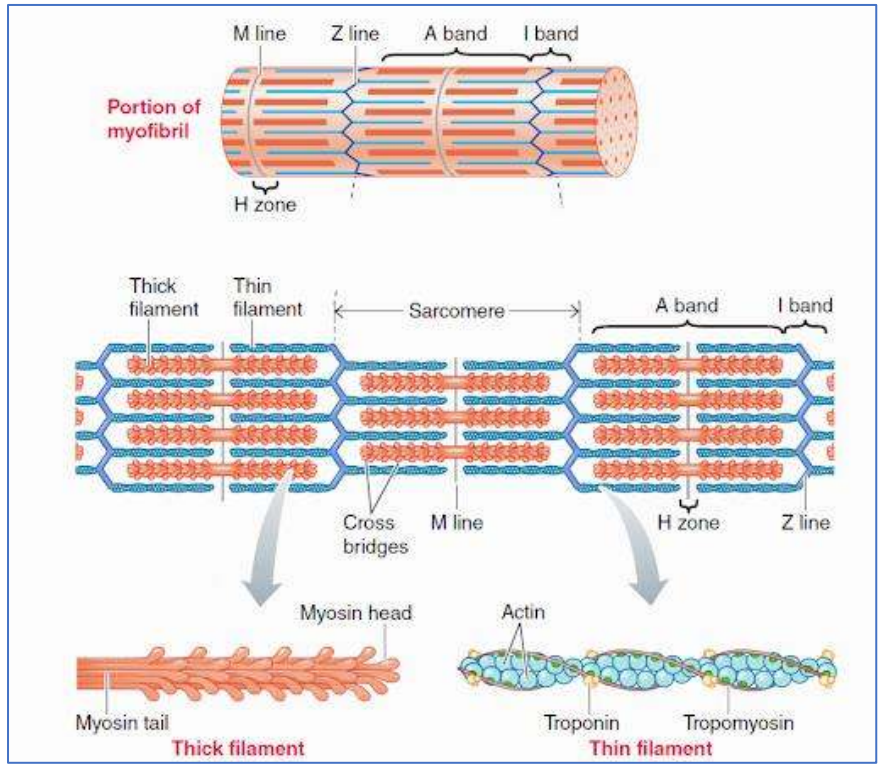


Figure 3.8 Calcium released from the sarcoplasmic reticulum binds to troponin C, triggering myosin–actin interaction through conformational changes in tropomyosin. Source: Used with permission from Hand *et al.*, 2010.

肌小節部位	肌肉收縮時的變化	為什麼？
A 帶 (A band / 暗帶)	✗ 不變	代表粗肌絲全長，粗肌絲本身沒有變短。
粗肌絲、細肌絲	✗ 不變	蛋白質纖維本身的長度固定不變。
I 帶 (I band / 明帶)	變短	僅含細肌絲的區域，隨細肌絲被往中央拉而縮短。
H 區 (H zone)	變短甚至消失	A 帶中央只含粗肌絲的區域，因細肌絲插入而縮短。
肌小節 (Sarcomere)	變短	兩條 Z 線（Z lines）被互相拉近，整個結構縮短。

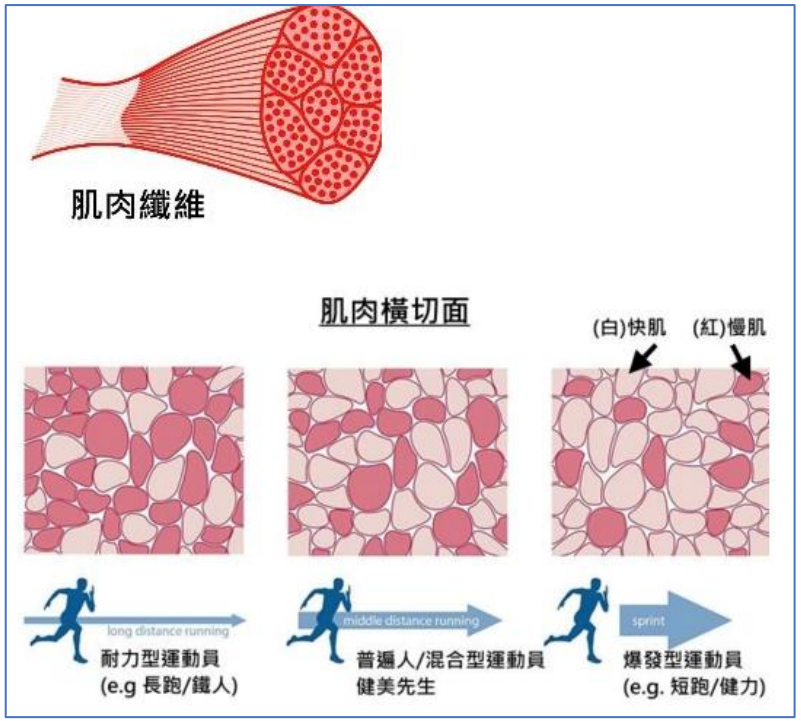


©Liu, Ching-Ming 2026

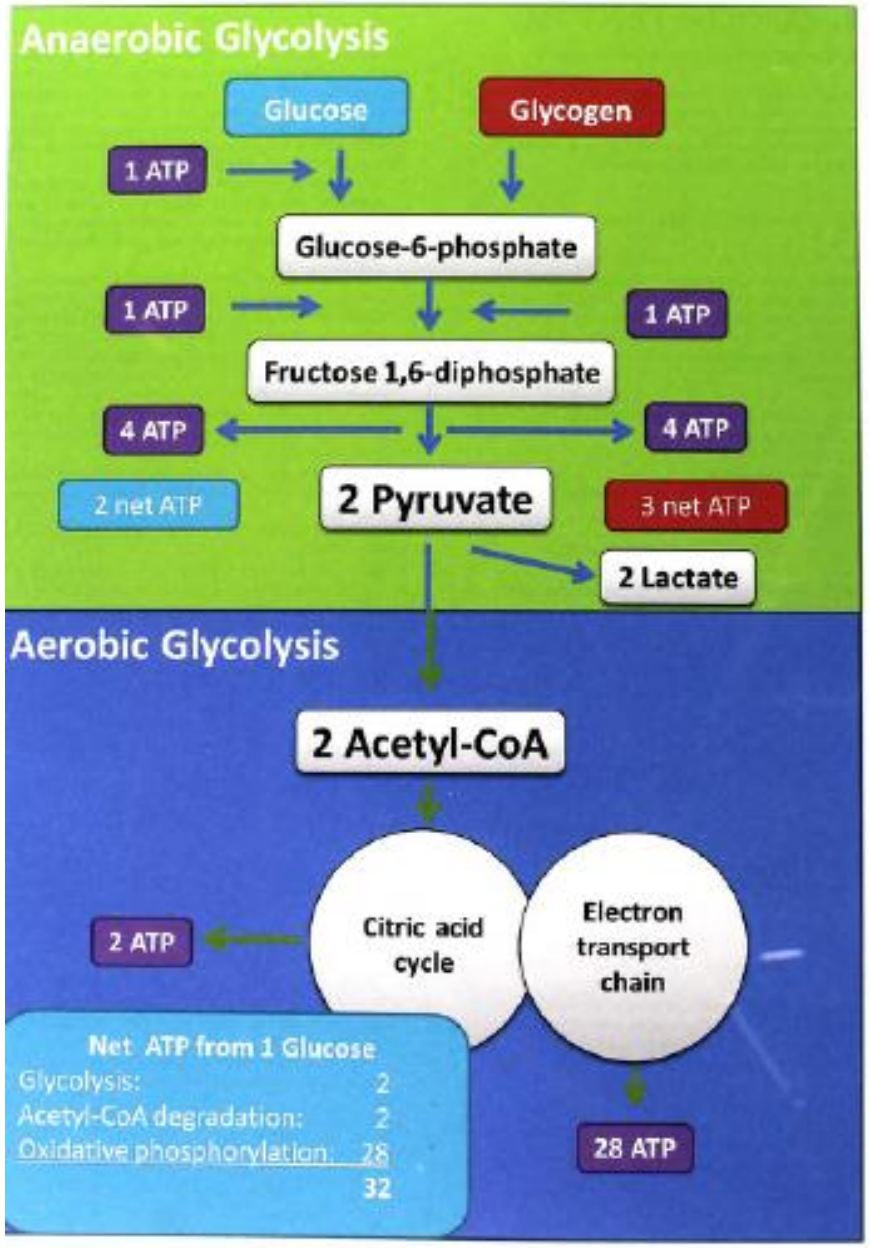
Muscle肌肉

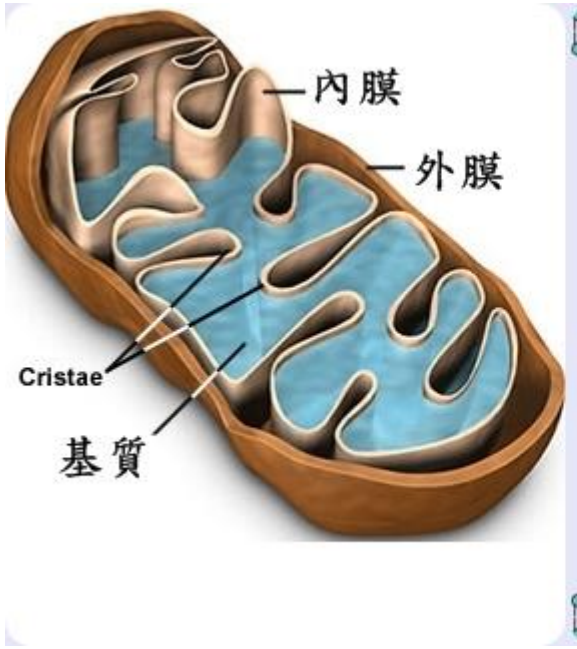
- Types of muscle fibers
  - Slow twitch fibers (Type 1), contain a larger quantity of oxidative enzymes, myoglobin and mitochondria, long contraction time, are resistance to fatigue, antigravity extensor contraction . They are well suited to perform aerobic activities. e.g. quadriceps
  - Fast twitch fibers (Type 2), contain large amounts of glycolytic enzymes for anaerobic metabolism, short contraction time, burst of strength, suitable for jumping, sprinting, powerful movement. Thus, they are able to quickly and powerfully contract and fatigue significantly faster than type I fibers. e.g. gastrocnemius
- 慢肌(第一型)為紅肌進行有氧代謝(氧化酶)有較高耐力並較能抗疲勞
- 快肌(第二型)為白肌進行無氧代謝(糖酵解酶)能提供最大力量和快速收縮，但容易疲勞

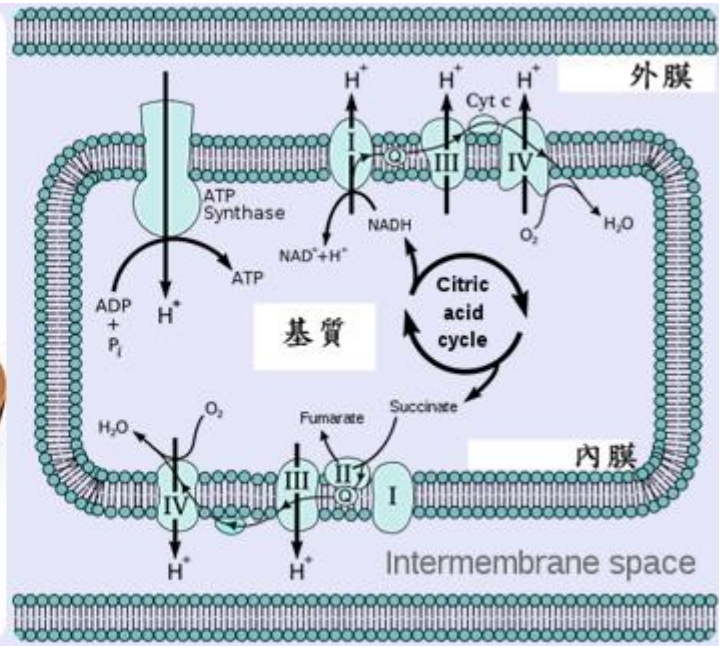
- In dogs: 5 types
  - types I, IIa and IIx
  - other two are hybrids (I + lla and IIax)
- The types of muscle fibres present in muscles are closely related to the function of individual muscles. It has been shown that the type of muscle fibres can change in conjunction with the muscle's function.
- 狗有5類肌纖維，肌肉纖維的組成型態與肌肉功能有關，隨著功能需求而分佈比例不同



From google





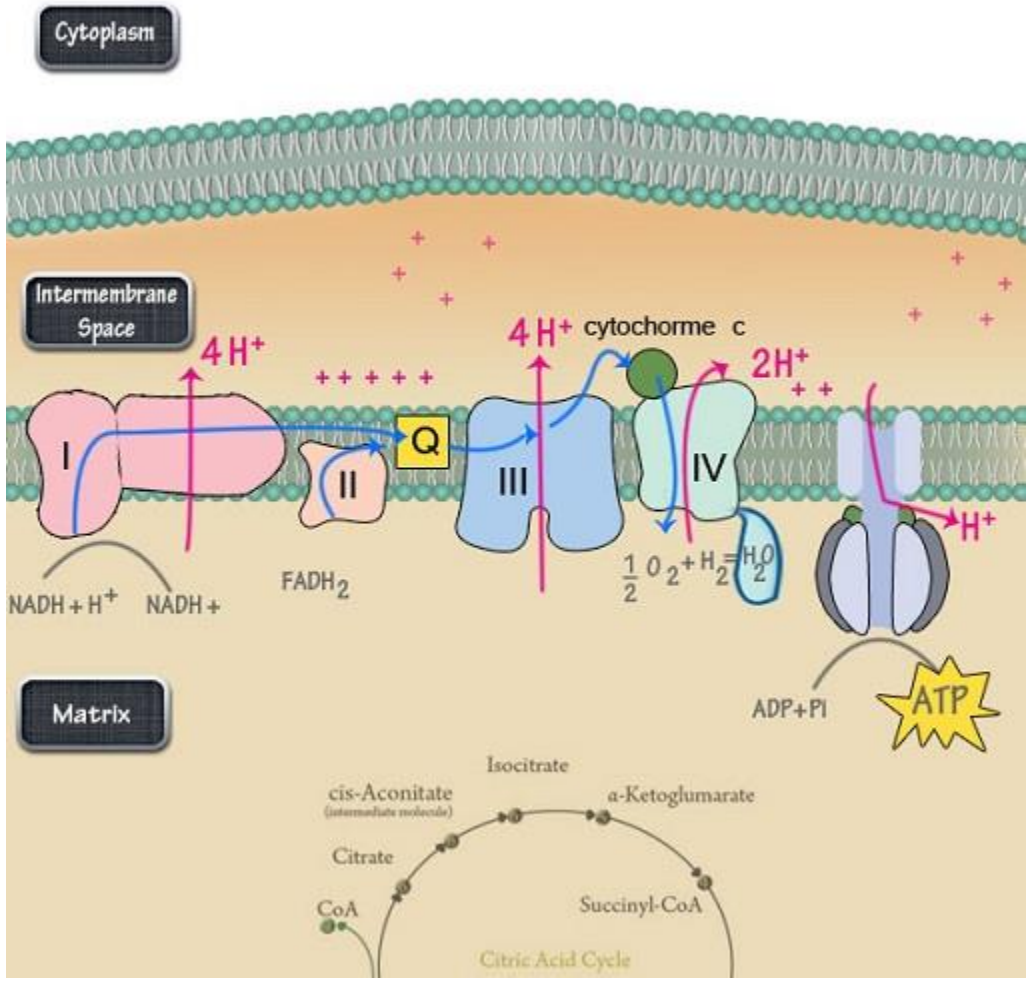


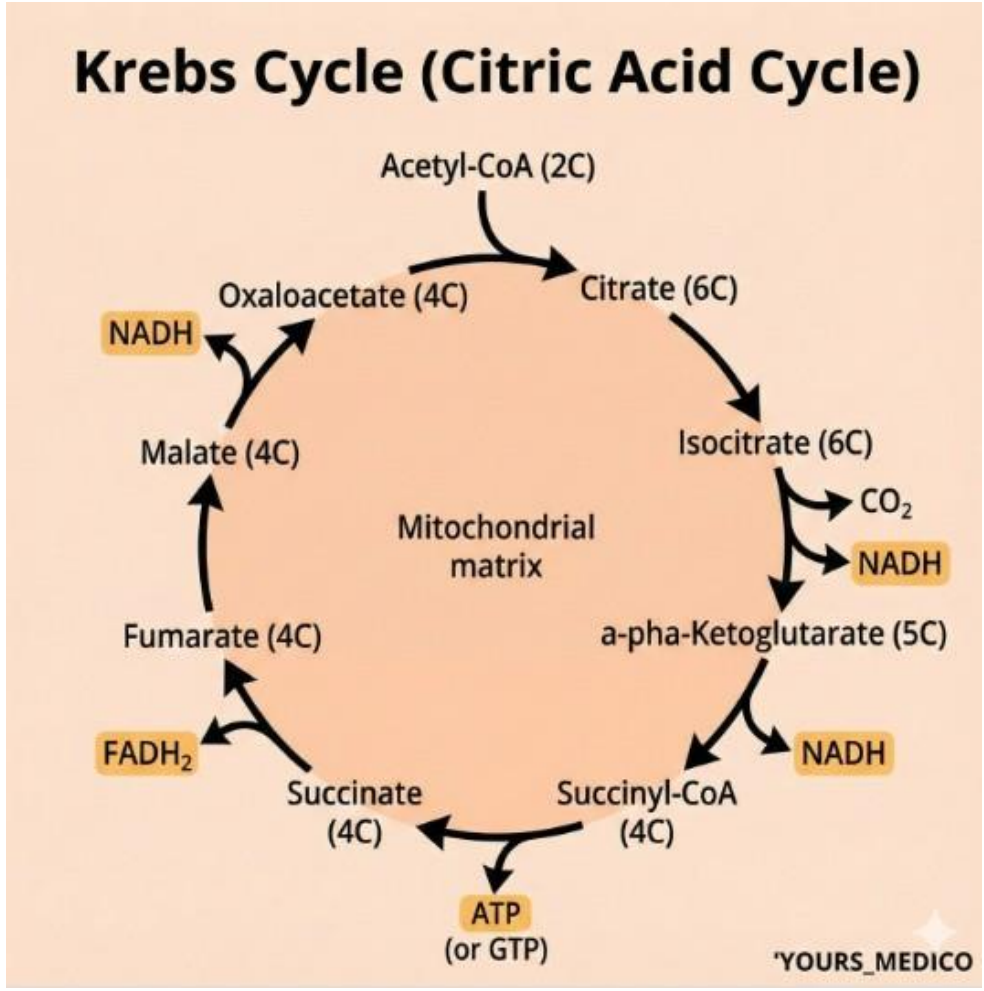
https://www.google.com/

能量系統	主要供能方式	供能時間	是否需要氧氣	關鍵字
即時能量系統	儲存的高能量物質直接提供能量	約 5～10 秒；頂尖運動員可達約 20 秒	✗ 不需要	最快、短時間、高強度
糖解系統	無氧分解葡萄糖產生能量	約 5～20 秒至 2 分鐘	✗ 不需要	無氧、葡萄糖、短中時間
氧化系統	透過有氧代謝／氧化作用產生能量	約 2 分鐘後開始成為重要供能來源	☑ 需要	有氧、氧化、長時間運動

Essential facts of physical medicine, rehabilitation and sports medicine in companion animals, Bockstahler 2019

©Liu, Ching-Ming 2026





https://www.google.com/

核心差異對比

比較項目	無氧醣解 (Anaerobic Glycolysis)	有氧醣解 (Aerobic Glycolysis)
氧氣需求	**不需要**氧氣	**需要**充分的氧氣
最終產物	**乳酸**（Lactate）與 2~3 個 ATP	**丙酮酸**進入粒線體，最終轉為 **二氧化碳、水**與約 30~32 個 ATP
能量產生速度	**極快**，能瞬間爆發能量	**較慢**，需要複雜的化學反應流程
能量產生總量	**極少**（每分子葡萄糖僅能產生 2~3 ATP）	**極多**（每分子葡萄糖可產生高達 30+ ATP）
運動強度	**高強度**運動（如 400 公尺全力衝刺、高強度間歇）	**中低強度**運動（如慢跑、健走、長距離騎車）
維持時間	短暫，約 **30 秒至 2 分鐘**	持久，可維持 **數十分鐘至數小時**
主要限制因素	乳酸與氫離子累積導致**肌肉酸痛、疲勞**	體內**肝醣耗盡**（即俗稱的「撞牆期」）

©Liu, Ching-Ming 2026

Muscle healing 肌肉癒合

- Types of muscle injuries
  - Contusion 挫傷(blunt)
  - Strains 拉傷(indirect)
  - Lacerations 撕裂傷(sharp)
- The severity
  - Stage I- myositis 肌炎
  - Stage II- fascial sheath inflammation 筋膜鞘發炎
  - Stage III- myofiber disruptions and hematoma 肌纖維斷裂及血腫
- Following the general wound healing- inflammatory, reparative, remodeling
- 依循一般傷口癒合模式–炎症期，修補期，重建期
- Rehab notes:
  - 0-5th day immobilization
  - Then controlled, protective movement for 4 weeks
  - After that, start intense exercises
- 復健提示：先期5天內固定，接著有限制地活動4週，之後才加強運動

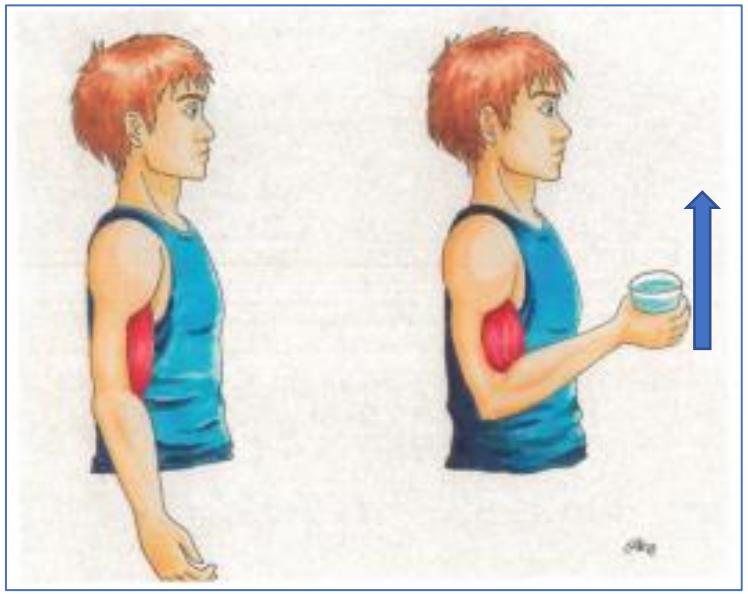
©Liu, Ching-Ming 2026

Muscle atrophy 肌肉萎縮

- **Disuse atrophy廢用性萎縮**
  - e.g. lameness or immobilization of a joint (bandage or external fixator)
  - Develop slowly, is reversible and incomplete
  - Extensors more susceptible than flexors
  - Postural muscles, mainly type I fiber
- 例如跛足或固定肢體，萎縮發展緩慢，大多屬伸肌群，維持姿勢的第一型肌纖維肌肉群
- **Neurogenic atrophy神經性萎縮**
  - e.g. paralyzed animal
  - Develop rapidly within 1 week, contracture followed by fibrosis
  - Depends on the severity of nerve damage
- 例如癱瘓動物，萎縮發展快速，約一週內，隨神經傷害嚴重程度不同，最嚴重會肌肉纖維化呈現攣縮現象

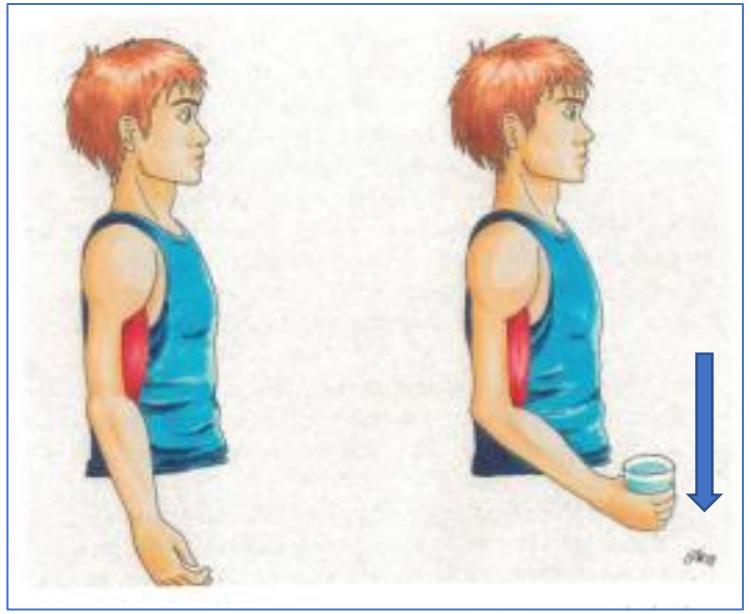
- **Sarcopenia肌少症**
  - Aging
  - Loss of myofibers due to apoptosis, decreases in myofiber size
  - Overall muscle circumference is often maintained during early sarcopenia, as lost myofibers are replaced with adipose and fibrous tissue
  - While it is impossible to fully prevent this from happening, there are two primary ways to slow the process in elderly patients, which include increasing dietary protein, and incorporating concentric exercises. e.g. limb weighting, aquatic therapy, and balance board exercises.
- 因老化肌纖維喪失，先以脂肪組織取代肌肉組織接著肌肉纖維化
- 不易逆轉或預防，兩種方式可以延緩，補充蛋白質及向心收縮運動，例如水中跑步機及平衡板運動
- Remobilization of muscles
  - Conditions of animal
  - Mechanical load
  - Intense training like treadmill, swimming
  - Eccentric 離心收縮, concentric 向心收縮 exercises
- 肌肉復健要點
- 依動物狀態做選擇，增加機械負荷，水療包括游泳，向心收縮及離心收縮運動

©Liu, Ching-Ming 2026



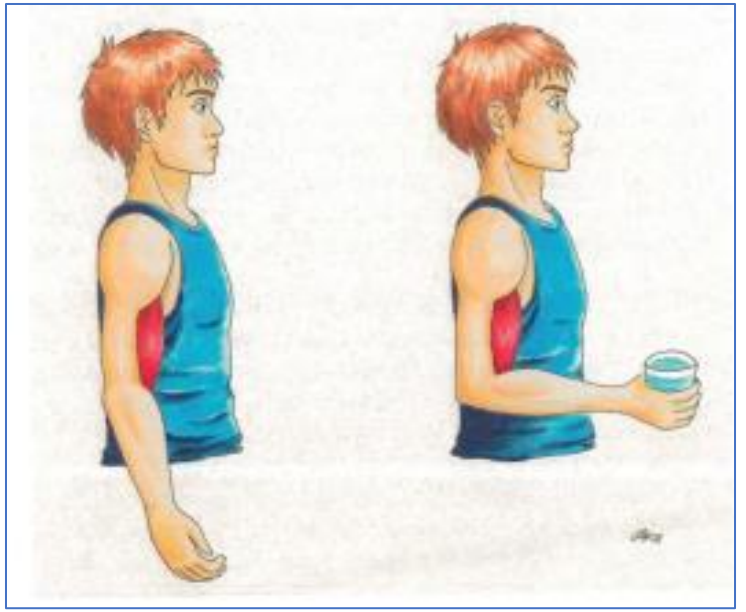
Concentric contraction

向心收縮



Eccentric contraction

離心收縮

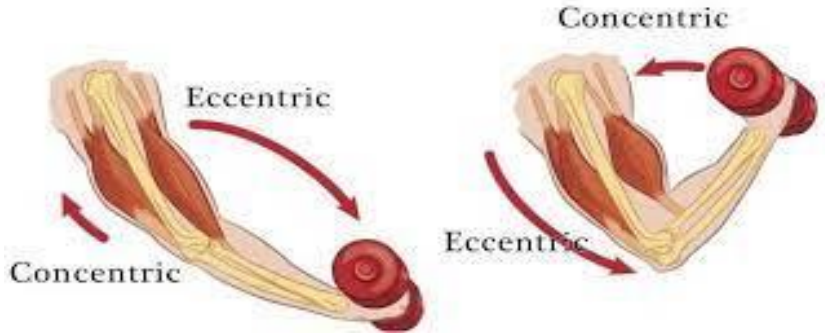


Isometric contraction

等長收縮

From google

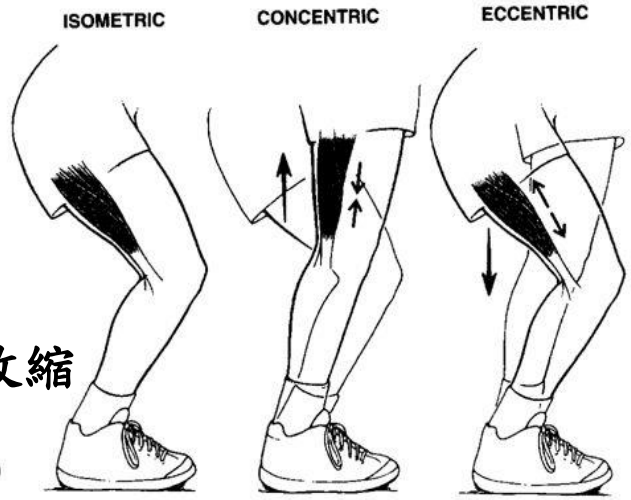
©Liu, Ching-Ming 2026

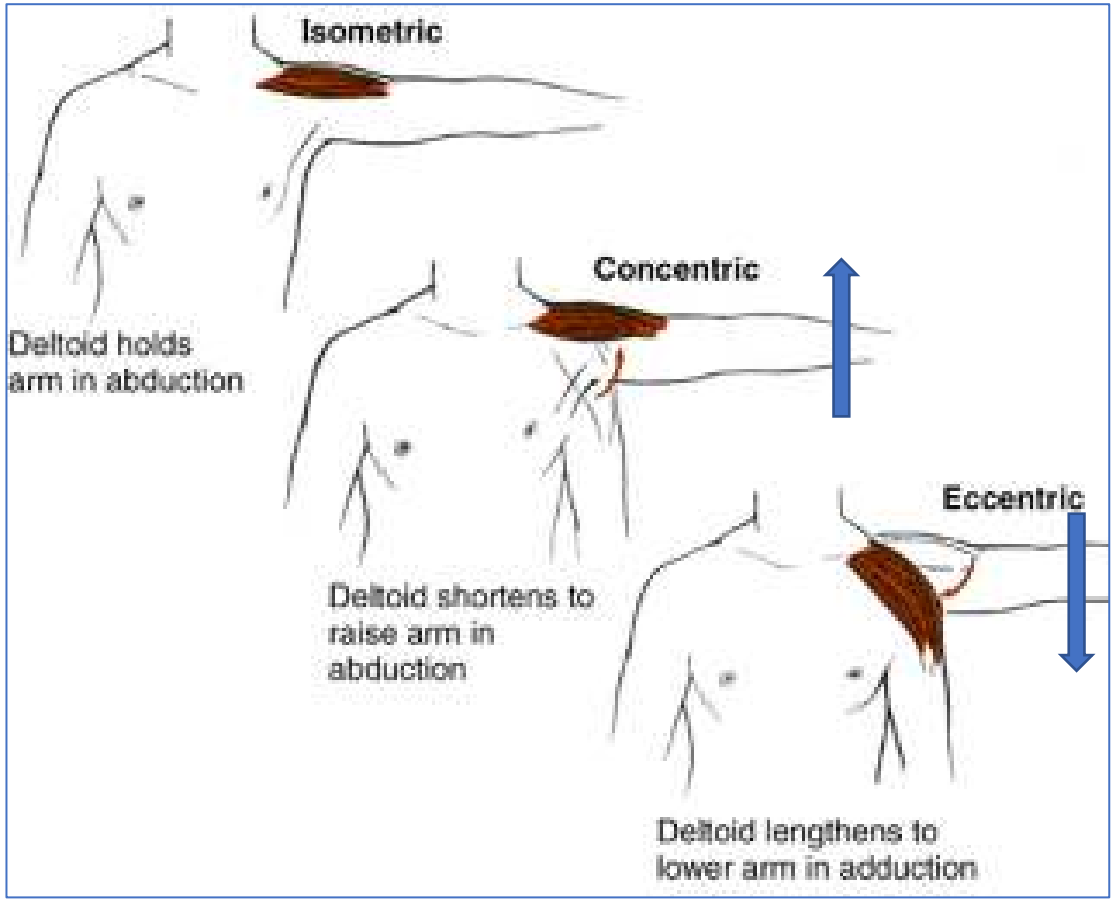


Muscular Action

Whole Muscle

- **isometric action**等長收縮
  - no change in fiber length
- **concentric action**向心收縮
  - shortening of fibers to cause movement at a jt
- **eccentric action**離心收縮
  - lengthening of fibers to control or resist a movement





©Liu, Ching-Ming 2026

Healing Timeline: Skeletal Muscle 骨骼肌

- Inflammation
  - Immediate to 72 hours (inflammation)
  - **炎症期達72小時**
- Repair
  - 3 days (type III collagen) for scaffold for myofibers; normal collagen ratio > 6 weeks
  - **修補期3天主要以第3型膠原蛋白為肌纖維支架，6週後才有正常比例膠原蛋白**
- Remodeling/Maturation
  - Ratio of functional muscle fiber to scar tissue depends greatly on size of gap at site of injury
  - **重建期形成結痂組織視受傷處大小而定**
- Healing/Strength Characteristics
  - Normal tissue regeneration is required for return to full function; occurs in minimum 4-6 wk.
  - Complete muscle lacerations near mid-belly in rabbits showed final return to 50% of tension-producing ability and 80% of normal contraction shortening at 12 wk.
  - Partial tears have an improved outcome for tensile strength; return to 60% of tensile strength and normal contractile ability achieved with partial lacerations at 12 wk.
  - **正常組織完全回復功能需6週，肌肉完全撕裂經12週復原達50%張力及80%收縮力，部分撕裂復原則經12週達60%張力及正常收縮力(兔子試驗)**

©Liu, Ching-Ming 2026

Implications for Rehabilitation of muscle

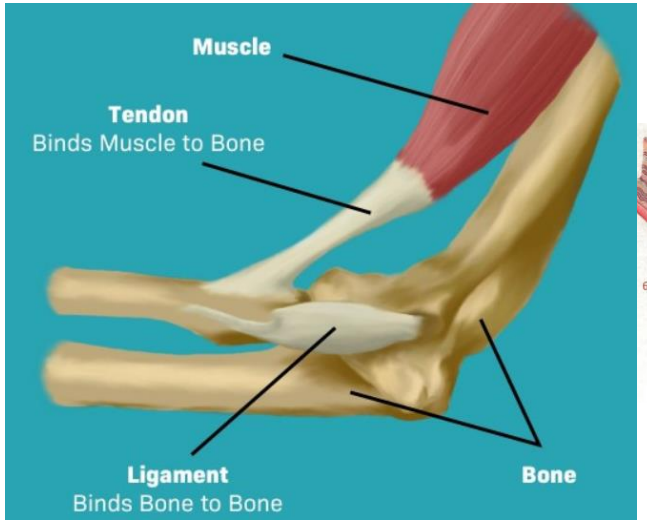
- Decreases in muscle contraction strength directly related to degree of injury and ingrowth of fibrous tissue (e.g. iliopsoas injury, infraspinatus contracture)
- 肌肉復原時收縮力降低與受傷程度及纖維組織取代程度直接相關

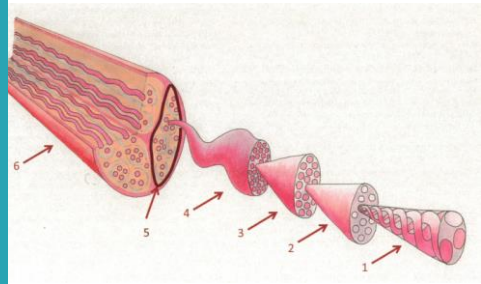
- Rehabilitation depends on extent of injury (i.e., degree of strain, rupture)
  - In mild strains, brief (3-5 day) immobilization followed by active muscle contractions in mobilization demonstrates better penetration of muscle fibers through the scar tissue and better muscle strength and size
  - For muscle tears, immobilize for a minimum of 2-3 weeks to protect repair, promote new collagen, and sustain remobilization stresses
  - For rehabilitation of complete rupture, moderate activity can begin at 4-8 weeks. At 8-12 weeks, unrestricted activity may be allowed
- 輕微拉傷，簡易固定3-5天後在活動肌肉可以讓肌肉有良好強度，
- 肌肉撕裂則需2-3週固定及保護
- 肌纖維完全斷裂者，4-8週才可開始中度活動，8-12週才能不限制活動

©Liu, Ching-Ming 2026

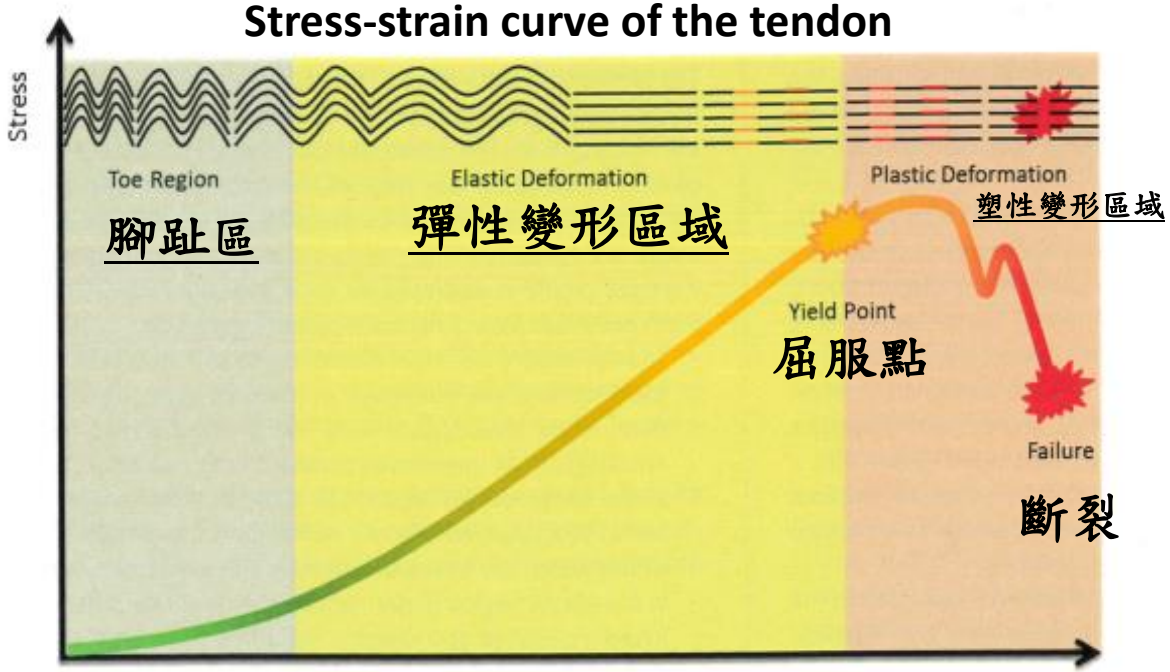
Tendon and ligament 肌腱與韌帶

- **Tendon 肌腱**
  - Connect muscle to bone
  - For extension or flexion of joints
  - >98% type I collagen able to change in length by ~5%
  - Paratendon 腱旁組織covered (**vascular**) (e.g. gastrocnemius and triceps muscle tendons)
  - Synovial-sheathed 滑膜鞘(**avascular**) (e.g. digital flexor tendons)
  - Immobilization for 3-6 weeks after tendon Sx
  - After 6weeks, light weight-bearing exercises can begin
  - After one year, 80% of the strength capacities of the tendon is achieved
- 肌腱連接骨頭與肌肉以伸屈關節，其營養供應分2類，分別為由血管(腱旁組織)及非血管(滑膜鞘)途徑，損傷經手術處理後需固定3-6週，6週後始可做輕微負重運動，即便如此，1年後肌腱張力恢復僅達80%
- **Ligaments 韌帶**
  - Connect bone to bone
  - For control movement and providing stability to the joints
  - 90% type I/10% type III collagen
  - Inter-, intra-capsular (e.g. cruciate ligament)
  - Extra-capsular (e.g. ligaments located in the vertebral column)
  - Contain Nociceptors, Golgi receptors
  - The healing and rehabilitation parameters are similar to tendons
- 韌帶連接骨頭與骨頭以穩定關節，分關節內及關節外2種，含Golgi感受體以偵測拉力承受強度，癒合與復健要點與肌腱相似





肌腱應力-應變曲線



Essential facts of physical medicine, rehabilitation and sports medicine in companion animals, Bockstahler 2019

©Liu, Ching-Ming 2026

Healing timeline of Tendon 肌腱癒合時間軸

- Inflammation
  - Immediate to 72 hours
  - **炎症期72小時**
- Repair
  - Day 5: random collagen orientation, progressively increasing for 4 weeks; collagen production peaks 5-12 days after injury, markedly decreased by day 60
- **修補期膠原纖維從第5天開始增加達4週，5-12天達高峰，60天才減少增加**
- Remodeling/Maturation
  - Takes a minimum 28 days for collagen fibers to align according to stress; tendons appear normal by 112 days
- **重建期至少28天膠原纖維依應力排列，112天肌鍵才呈現正常**
- Healing/Strength Characteristics
  - At 6 weeks, 56% normal tensile strength occurs, 79% at 1 year and beyond. Surgical repair of fresh gap required for appropriate organization of scar at 6 weeks.
  - Tendons with gaps > 3 mm have lower tensile strength at 6 weeks compared to those with smaller gaps.
  - Synovial (avascular) tendons require longer healing times than vascular tendons.
- **肌腱強度6週達56%，1年以上才達79%強度，手術方式修復肌腱需6週才結痂，經6週修復，肌腱斷裂間隔超過3mm者期肌腱強度比小於3mm恢復差**

©Liu, Ching-Ming 2026

Implications for Rehabilitation of tendon

- Post-repair support for larger tendons may be needed for up to 6 weeks
  - Method and extent varies depending on surgeon preference, location of the injury and patient behavior
- Prolonged immobilization may be detrimental
  - Complete immobilization for longer than 21 days results in reduced vascularization at the wound site
  - Early controlled passive motion beginning within 21 days of repair appears to be ideal
- With full or partial transection and surgical repair, passive mobilization exercises appear to be optimal to begin at day 5 following surgery
- 大肌腱修復需至少6週時間，長期固定肌腱是有害的，固定肌腱超過21天會減少血管新生，21天內早期有限制地被動活動有益肌腱修復，最好於手術後第5天就開始有限制地被動活動
- Conservative exercise with light weightbearing may begin at 6 weeks, because having 25-30% of normal tensile strength for tendons has been considered adequate for withstanding normal muscle force
  - Minimum 12 weeks for less restrained, more active loading
- 6週就可以開始輕微負重之保守運動，因為25-30%肌腱強度足以負載肌肉力量
- Light exercise to maintain ROM and stimulate healing callus cells
  - Some weightbearing is required for maturation of callus
- 輕微運動以維持關節活動可以刺激甚至需要一些負重來增進癒傷組織細胞癒合
- Sufficient tensile strength return for normal activity load bearing **may take months**
- 肌腱恢復正常強度足以正常負重活動需好幾個月

©Liu, Ching-Ming 2026

Healing Timeline: Ligament 韌帶(Sprain 扭傷)

- Inflammation
  - Immediate to 72 hours
  - **炎症期72小時**
- Repair
  - 2-3 days to 6 weeks
  - **修補期2-3天至6週**
- Healing/Strength Characteristics
  - Occurs in several days to more than 1 year depending on grade (I-III)
  - At 1 year, 50-70% tensile strength occurs
  - Healing properties and rate of healing depend on which ligaments are involved
  - Lengthening may occur because of gap formation
- **韌帶癒合須數天至超過1年，其強度1年才達50-70%，因不同韌帶而異**

©Liu, Ching-Ming 2026

Implications for Rehabilitation of ligament

- Time to begin rehabilitation varies widely depending on the ligament involved.
- When in doubt, parameters for tendon healing may generally be followed
- 何時開始韌帶的復健因不同韌帶而異，基本上跟肌腱類似

- Tensile strength reduction may be slowed or decreased with joint motion during immobilization and stress reduction
- 關節活動減少及應力減少，韌帶強度及張力也隨之減弱

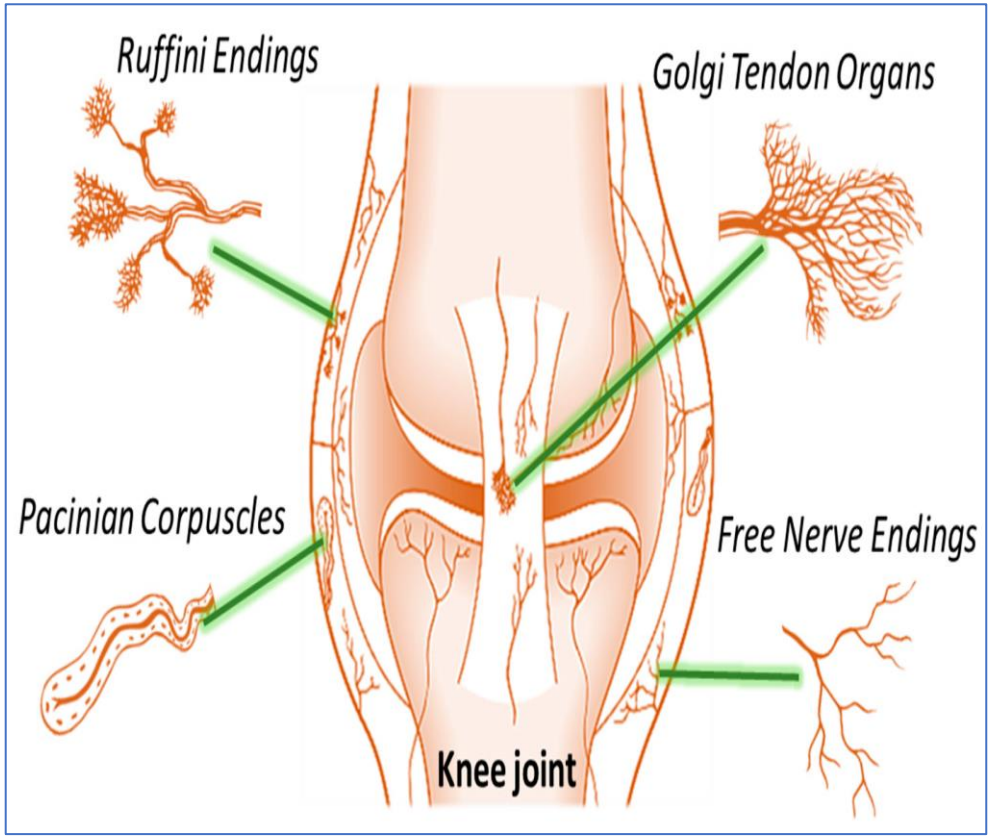
- Low-duration (30 min) and high frequency (6 days/week) exercise appears to have greatest benefit to ligament strength from one study in horse
- 在馬的一項研究，每次30分鐘運動 ，1週6天，有益韌帶強度

- Sprain-avulsion fractures at ligament-bone interface have best prognosis for return to normal tensile strength
- 韌帶骨頭連接點一起撕裂骨折有較佳的回復預後

©Liu, Ching-Ming 2026

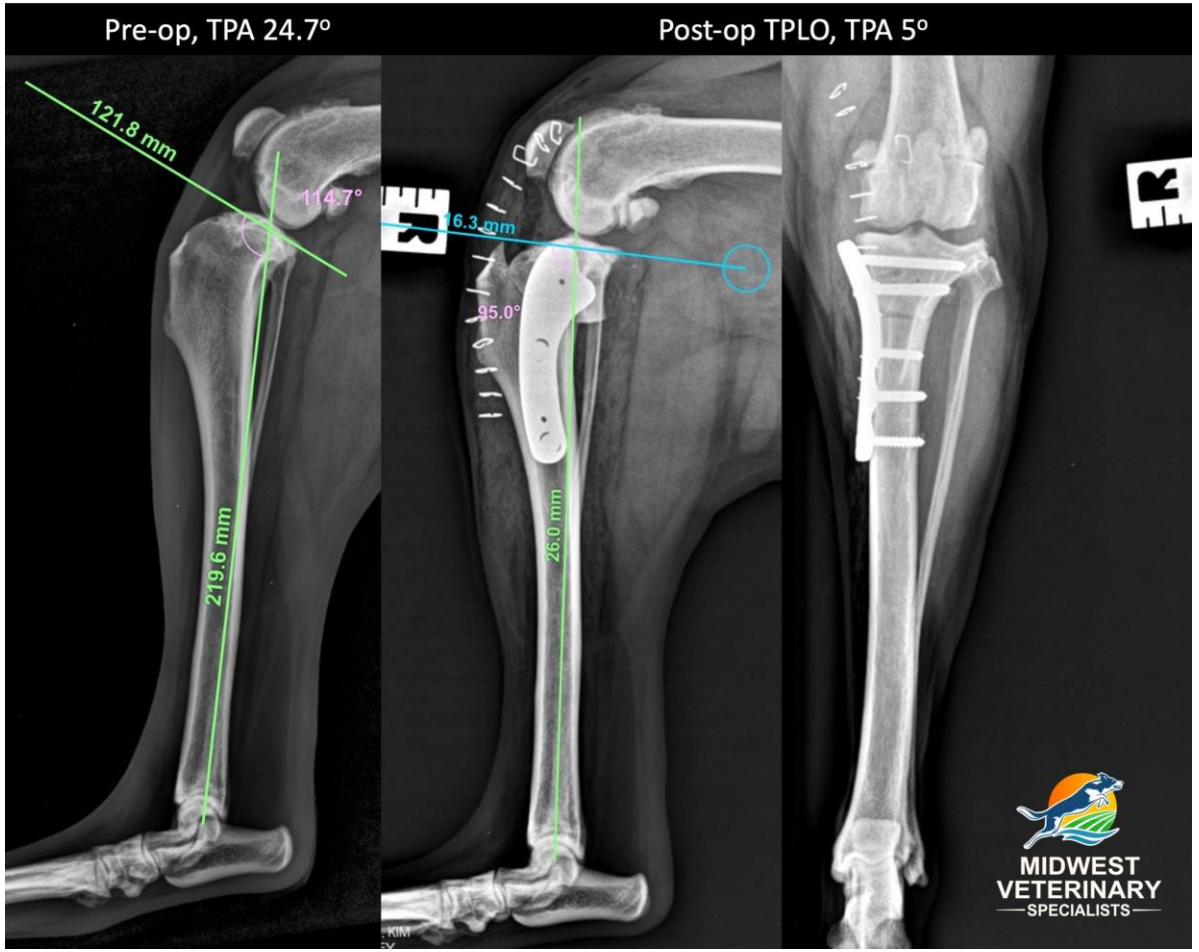
Joint capsule 關節囊膜

- The joint capsule surrounds the joint and produces synovial fluid.
  - Allows for frictionless movement
  - The synovial fluid provides the cartilage with nourishment.
  - Many proprioceptive receptors are embedded in the joint capsule. These receptors provide "position of the body in space" information.
- 關節囊包覆關節保護及產生關節囊液以潤滑關節，並含感受體偵測體位及運動動作
- The capsule surrounding the joint consists of:
  - An outer fibrous layer (primarily collagen) and serves as mechanical stabilization
  - An inner synovial layer of synovial cells closely related to the sub-synovial vascular plexus
  - The inner layer consists of synoviocytes type A (phagocytic activity) and type B (production of hyaluronic acid).
- 關節囊外層為纖維組織，其作用為力學穩固，內層含血管叢有A型滑液膜細胞具吞噬作用，B型滑液膜細胞，分泌玻尿酸潤滑用
- The capsule contains nociceptors, Pacinion Corpuscles and Ruffini Endings.
  - The ruffini bodies (joint receptors type I) transmit information about the position and movement of the joint.
  - The pacini-bodies (joint receptors type II) transmit information about joint movements and are therefore also called dynamic receptors.
- 關節囊膜有2種感受體，分別為 **Ruffini Endings and Pacinion Corpuscles** ，
- **Ruffini 感知關節動作，Pacini 感知關節體位及關節動作**



©Liu, Ching-Ming 2026

- Fracture
- TPLO



©Liu, Ching-Ming 2026

- Iliopsoas strain (髂腰肌拉傷)
- Fibrotic myopathy (纖維化肌病)
  - Infraspinatus contracture (棘下肌攣縮)
  - Gracilis/semitendinosus myopathy (股薄肌/半腱肌病)

- Sarcopenia (肌少症)

©Liu, Ching-Ming 2026

- Iliopsoas strain (髂腰肌拉傷)
- Origin:
  - the iliacus muscle originates on the ventral side of the ileum
  - and the psoas major originates on the transverse process of L2-L3 and on the ventral vertebral bodies of L4-L7.
- Insertion:
  - the two muscles fuse into a common tendon and insert on the lesser trochanter of the femur.
- Function:
  - responsible for advancing the rear limb forward, a hip flexor, an external rotator of the hip and a stabilizer of the spine.

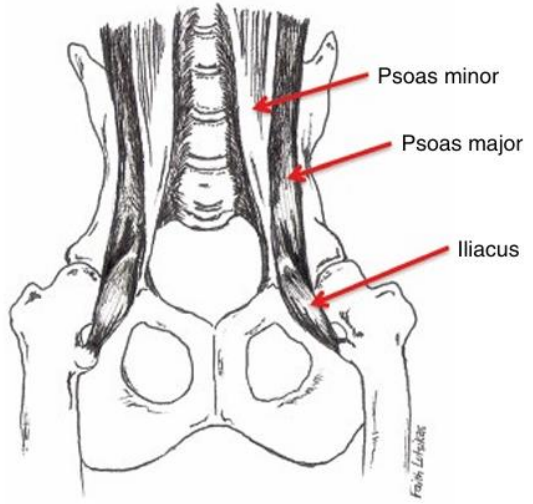


Figure 14.1 Iliopsoas muscle anatomy. The iliopsoas muscle represents the fusion of the psoas major and iliacus muscles. Source: Illustration by Dr. Faith Lotsikas.



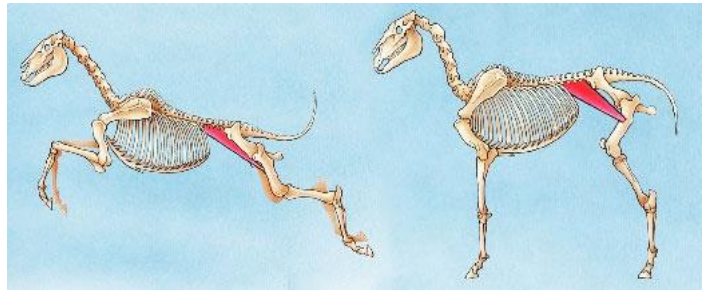


Table 1: Grading Scheme for Acute Musculotendinous Strain Injury

GRADE	DESCRIPTION
GRADE I	("Mild strain") no loss of function, <5% muscle involved, focal oedema or haemorrhage
GRADE I-II	As above but with mild fascial tearing
GRADE II	("Moderate strain") reduced strength of musculotendinous unit, >5% muscle involvement, mild fibre rupture, increased oedema/haemorrhage
GRADE II-III	As above but with fascial tearing, mild fibre disruption
GRADE III	("Severe strain") significant fascial tearing, significant to complete muscle fibre disruption, significant oedema/haemorrhage

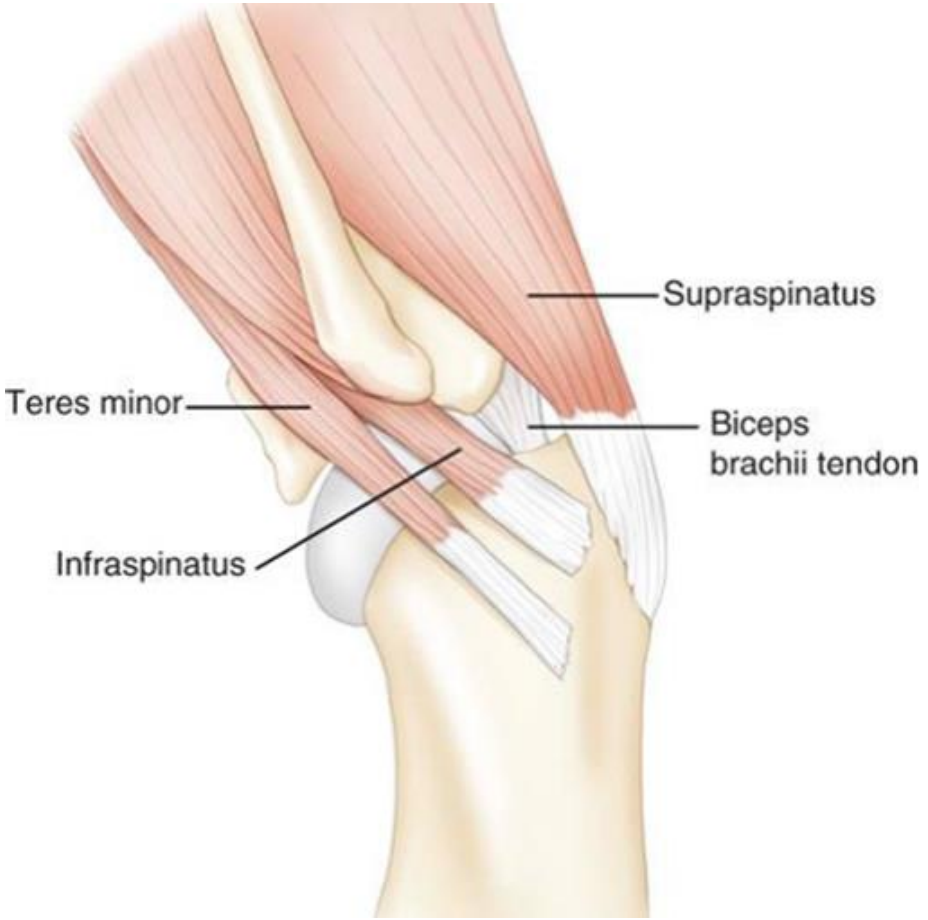
Veterinary Evidence , 2017 Vol 2, Issue 2 DOI http://dx.doi.org/10.18849/ve.v2i2.93

©Liu, Ching-Ming 2026

- Infraspinatus contracture (棘下肌攣縮)
  - Origin:
    - in the infraspinatus fossa of the scapula, courses laterally under the deltoid muscle
  - Insertion:
    - inserts on the greater tubercle.
  - Function:
    - It abducts and externally



https://www.google.com/



©Liu, Ching-Ming 2026

- Gracilis contracture（股薄肌孿縮）
  - Origin:
    - It originates on the pelvic symphysis.
  - Insertion:
    - inserts onto the cranial tibia but also into the crural fascia. From its caudal border, the aponeurosis sends a reinforcing band to the common calcaneal tendon with the semitendinosus.
  - Function:
    - the gracilis is a powerful adductor of the hip and can contribute to stifle extension.

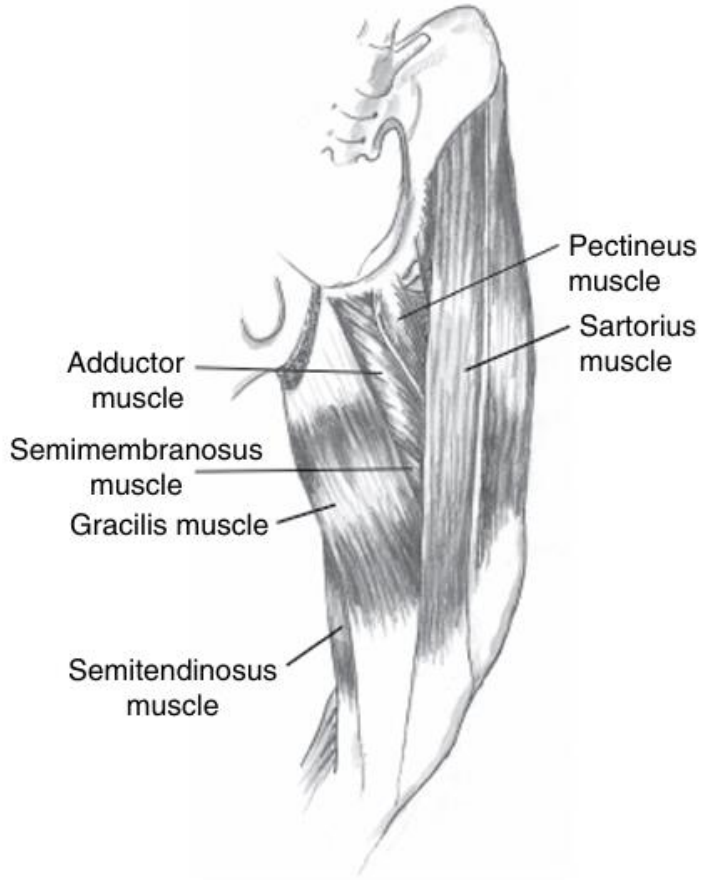
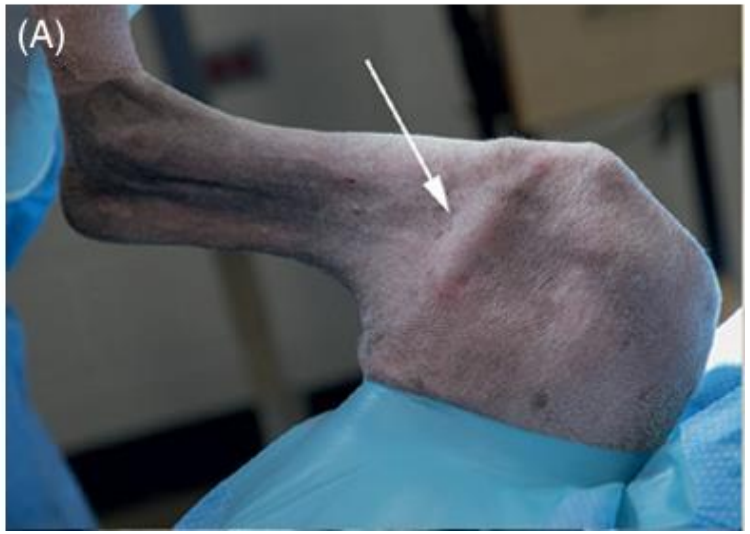


Figure 14.7 The gracilis, and lateral to that muscle, the semitendinosus and semimembranosus muscles, form an extensive broad muscular sheet that is found superficially in the caudal portion of the medial thigh. Source: Illustration by Marcia Schlehr.

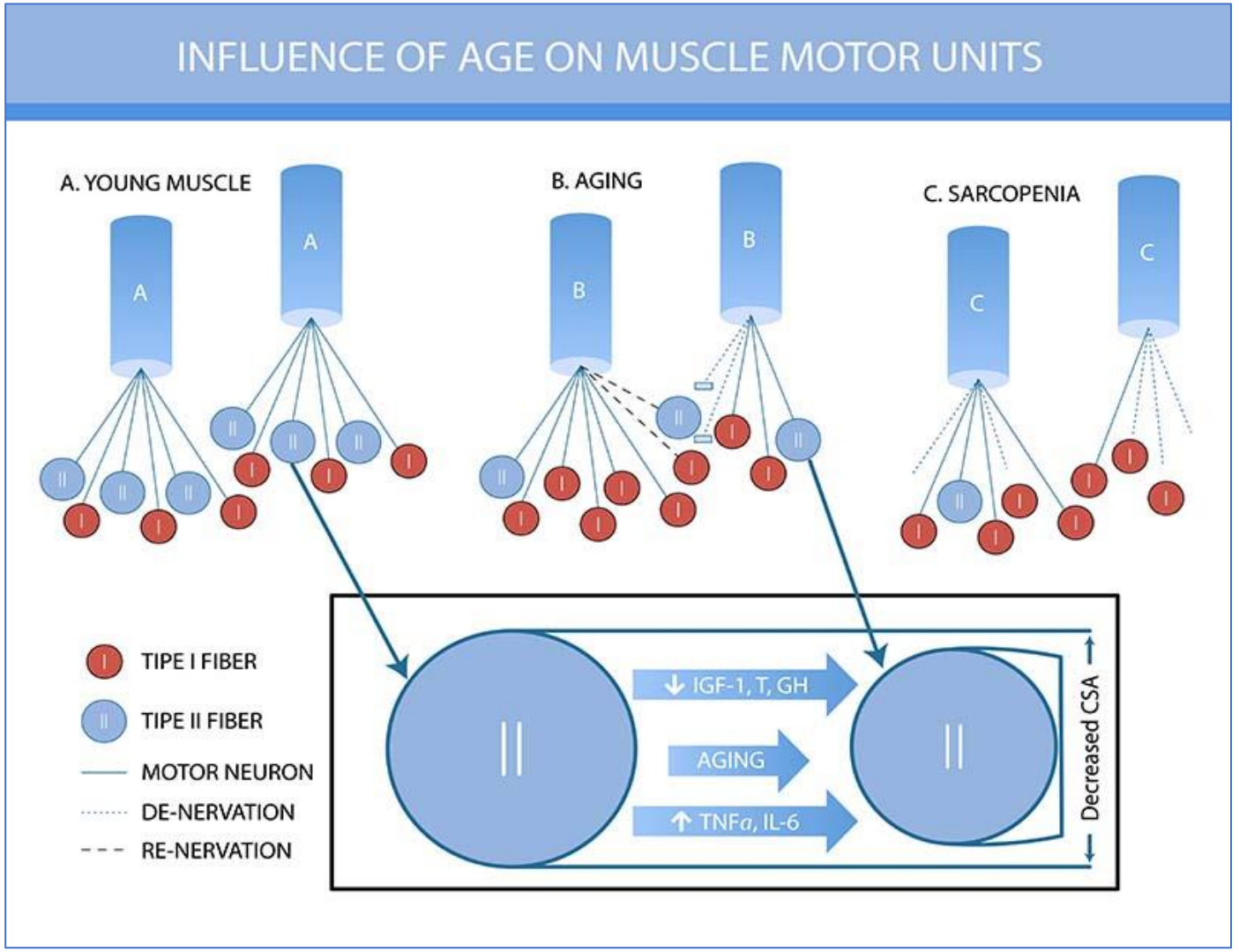


Canine Sports Medicine and Rehabilitation, 2nd Edition

©Liu, Ching-Ming 2026

- Sarcopenia肌少症

- Type I fiber
- Type II fiber

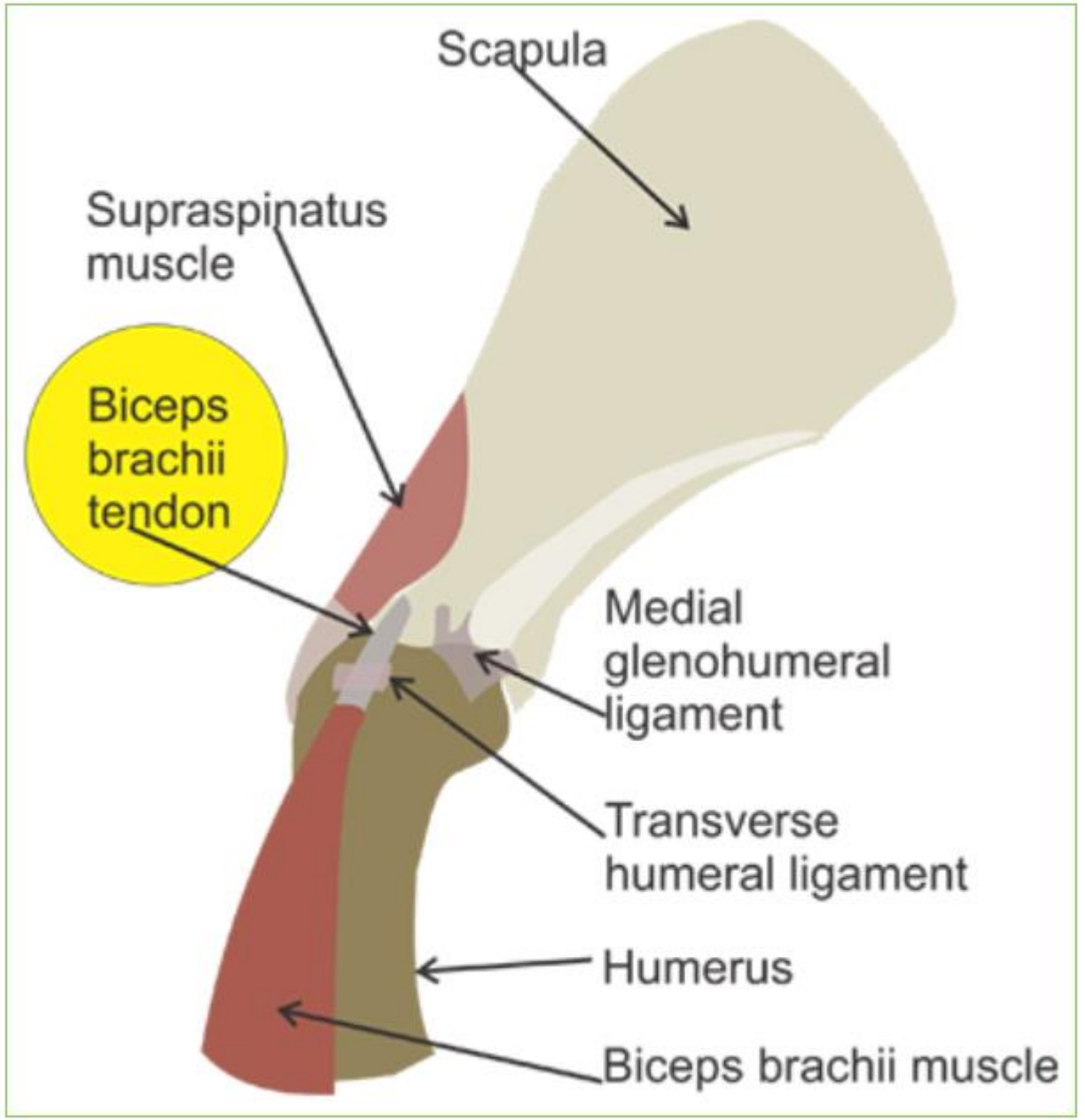


https://www.google.com/

©Liu, Ching-Ming 2026

- Supraspinatus tendinopathy
  - Origin:
    - in the supraspinous fossa of the scapula and extends laterally over the shoulder
  - Insertion:
    - attaching to the greater tubercle of the humerus
  - Function:
    - stabilizer and extensor of the shoulder joint

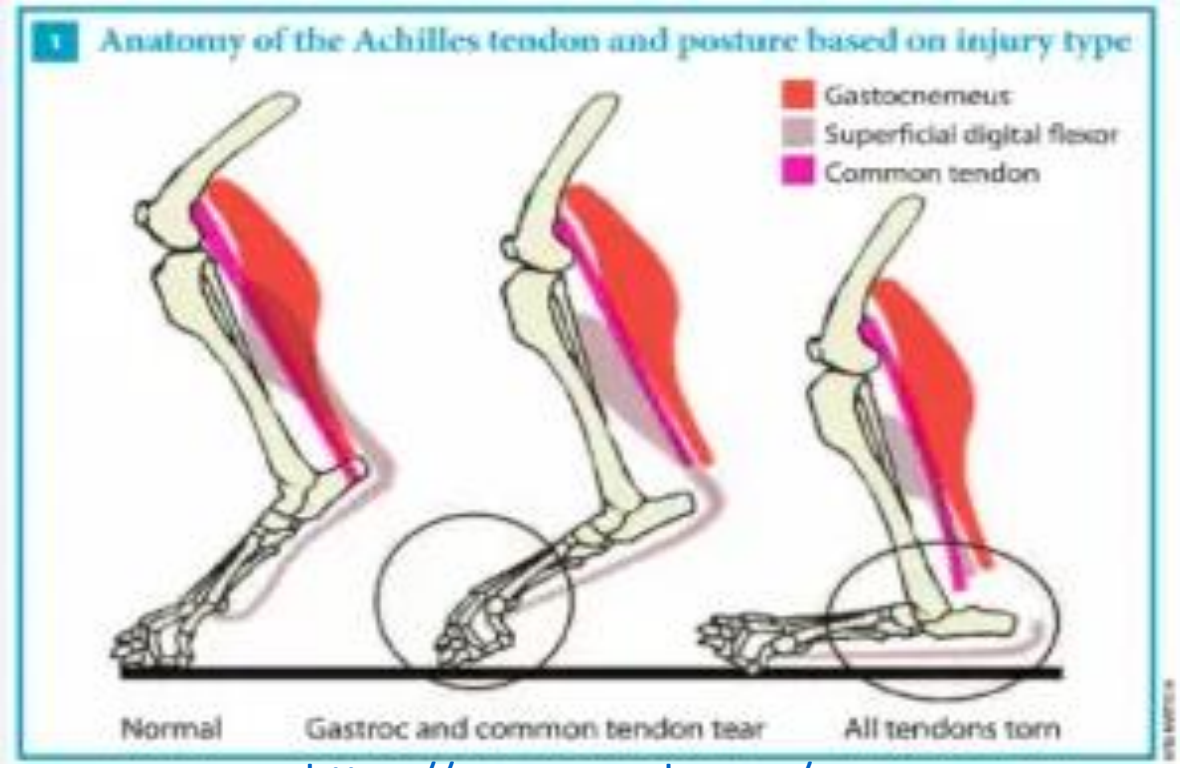
- Biceps tendinopathy
  - Origin:
    - on the supraglenoid tubercle of the scapula and courses cranial distally through the intertubercular groove
  - Insertion:
    - The muscle is located on the medial side of the humerus
    - inserts distally as a large, round tendon proximal to the elbow joint. The distal tendon bifurcates and the stronger of the two parts inserts cranial medially on the ulna and the weaker part inserts a little further distally on the radius
  - Function:
    - shoulder extension and elbow flexion



https://stemcellsafari.com/conditions-treated/inflammatory-disorders/biceps-tendonitis/

©Liu, Ching-Ming 2026

- Achilles tendon rupture (Dropped hock)
  - SDF (Superficial Digital Flexor) (淺趾屈肌)
  - Gastrocnemius (腓腸肌)
  - Cojoined tendon
    - Biceps Femoris（股二頭肌）
    - Semitendinosus（半腱狀肌）
    - Gracilis（股薄肌）



https://www.google.com/

Box 18.1 Grading of Pathology of the CCT (Meutstege 1993)

1) Type I = complete rupture
   (a) Stance = plantigrade
2) Type II = partial rupture
   (a) Stance = slightly increased hock flexion
      i) Type IIa affects the musculotendinous junction
      ii) Type IIb is similar to Type I but has an intact paratenon and therefore dogs are not completely plantigrade
      iii) Type IIc affects the GT only (with intact SDFT) resulting in increased flexor tension on the digits ("crab claw")
3) Type III = tendinosis only
   (a) Stance = normal

©Liu, Ching-Ming 2026

- SDF strain (Flat toe)

(SDF; Superficial Digital Flexor;淺指屈肌)

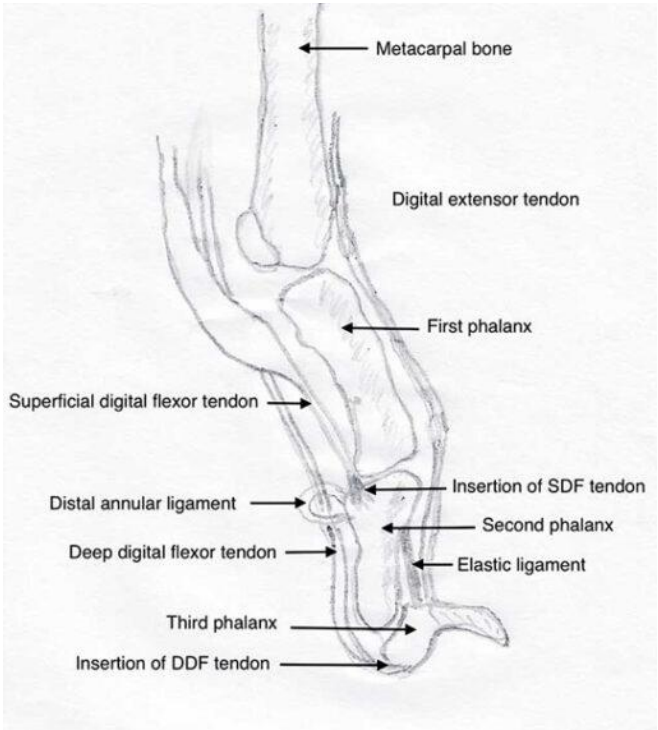
- DDF strain (Knucked up toe)

(DDF; Deep Digital Flexor;深指屈肌)



Figure 5 (left): The position of the digit after a superficial digital flexor tendonectomy. The P1/P2 joint is less flexed and is lower than the joint on adjacent digits to the ground. The digit is extended cranially, but the nail remains in contact with the ground. **Figure 6 (right):** The position of the digit after a combined tenotomy. The digit is flattened with a raised nail.

https://www.veterinarypracticenews.com/tendonectomy-offers-relief-for-chronic-corns-in-sighthounds/

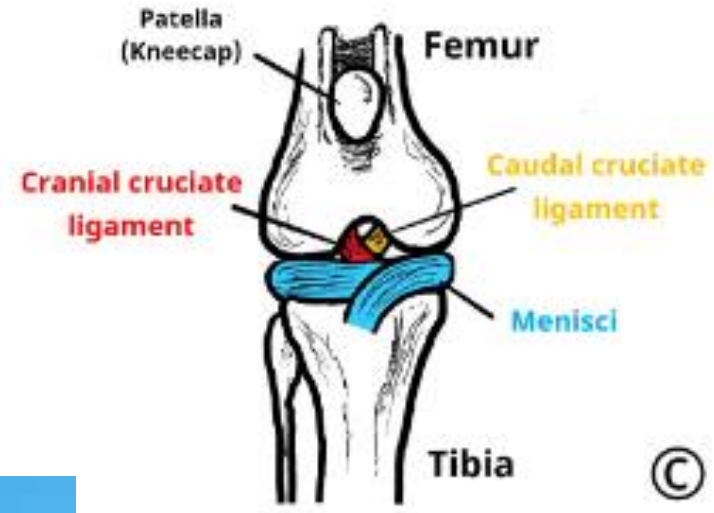


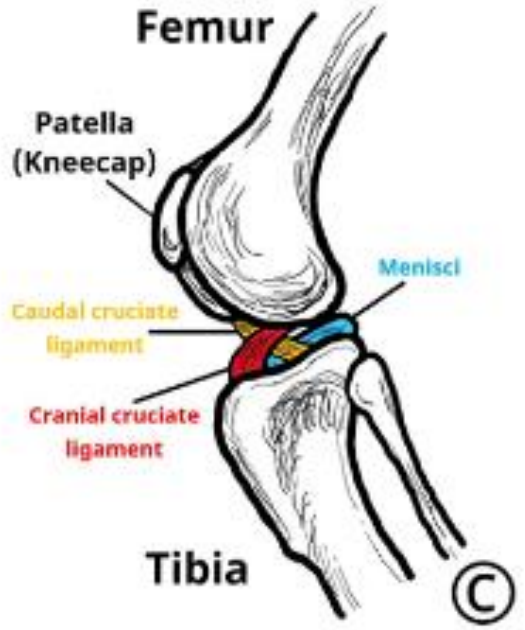


©Liu, Ching-Ming 2026

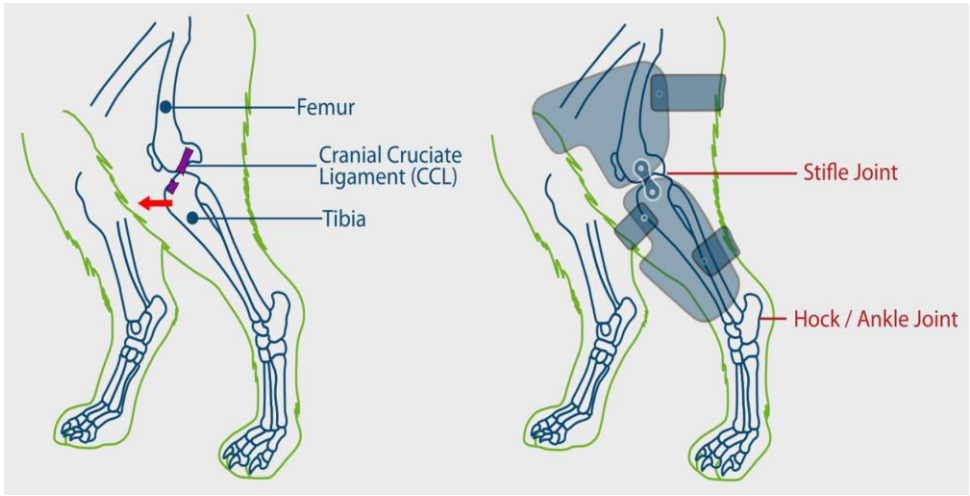
- CCLR (Cranial Cruciate Ligament Rupture)
  - Partial
  - Complete

  - Drawer test
  - Tibial thrust test









https://www.pawopedic.com/cranial-cruciate-ligament-tear

©Liu, Ching-Ming 2026

- MSI (Medial Shoulder Instability)
  - partial tearing of the MGL and SST
  - abduction test
  - median shoulder abduction angle of ~45° in clinically normal dogs

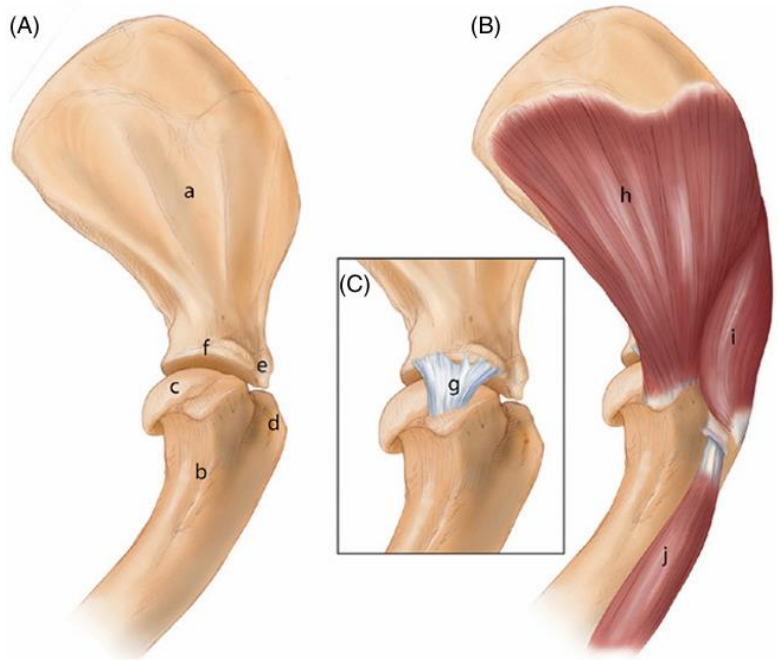


Figure 15.3 Normal anatomy of the medial shoulder for (A) osseous structures; (B) muscles; and (C) the collateral ligament of the medial shoulder: (a) scapula; (b) humerus; (c) humeral head; (d) greater tubercle; (e) supraglenoid tubercle; (f) glenoid; (g) medial glenohumeral ligament; (h) subscapularis; (i) supraspinatus; and (j) biceps brachii.



©Liu, Ching-Ming 2026

Joint and Cartilage

- Types of joint
  - Synovial joint (e.g. shoulder)
  - Fibrous joint (e.g. coronal suture)
  - Cartilaginous joint (e.g. pubic symphysis)
- 3 types of cartilage- hyaline, elastic, fibrocartilage
- Hyaline cartilage 透明軟骨
  - Minimize friction of movement within the joint
  - Absorbs forces acting on the joint

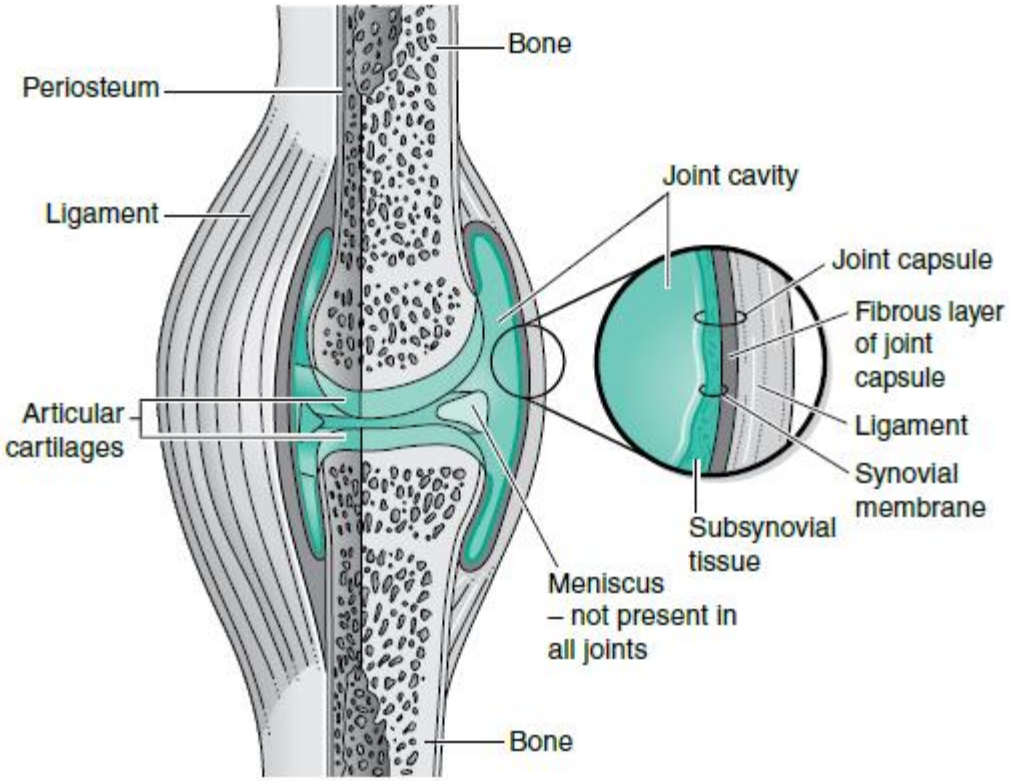


Fig. 3.24. Structure of a typical synovial joint.

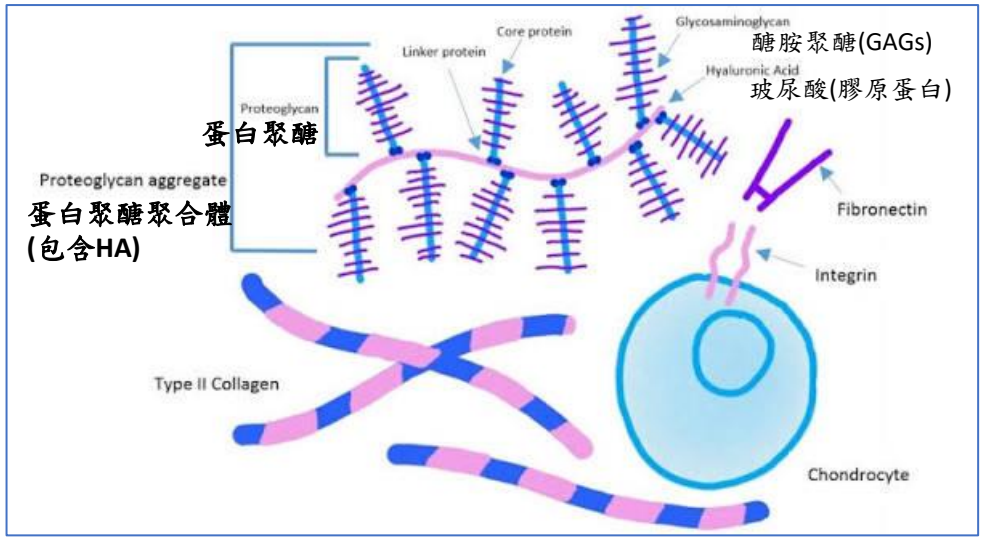
關節結構

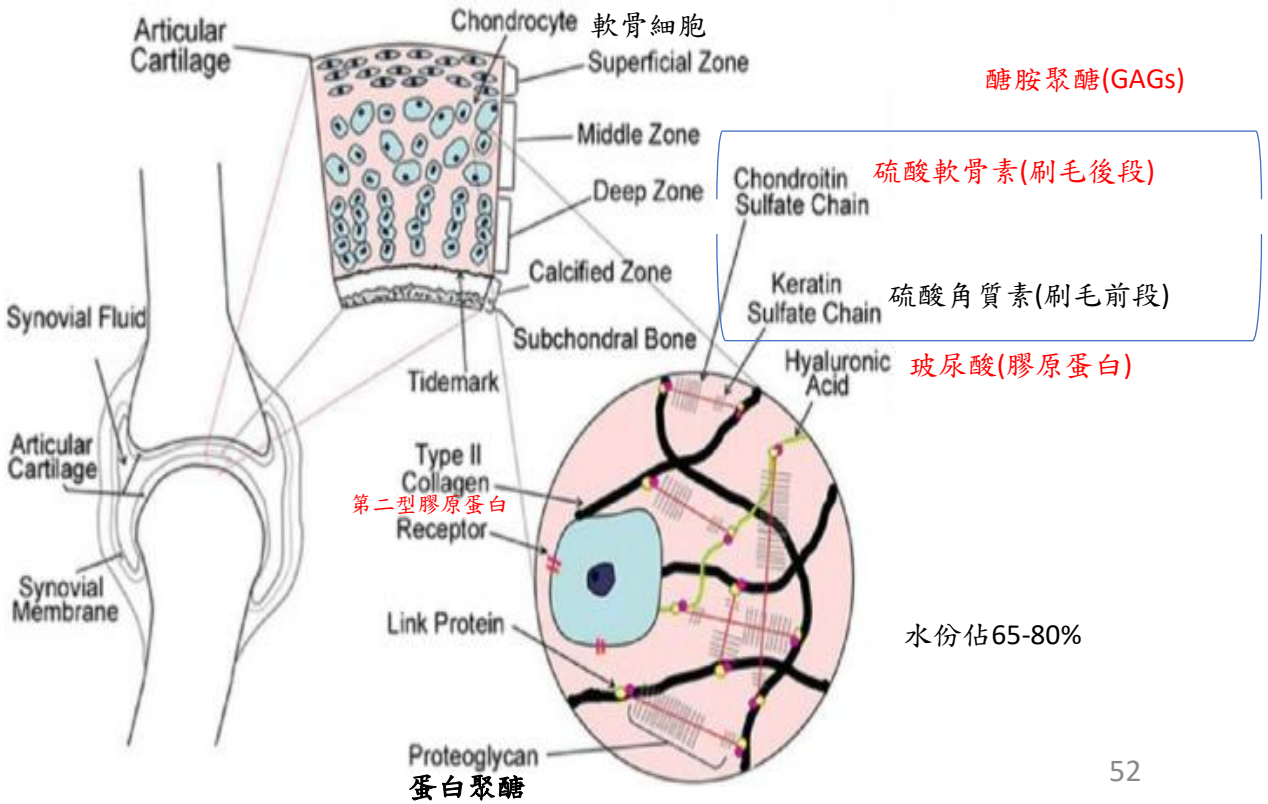
- Cartilage軟骨
  - Chondrocytes, extracellular matrix and water

  (軟骨細胞, 細胞外基質及水份)
  - No blood supply, no nerve supply, no lymph supply

  (無血液供應, 無神經分佈, 無淋巴分佈)
  - Nutrition and waste via synovial fluid exchange

  (營養及廢物完全靠關節囊液交換)





©Liu, Ching-Ming 2026

Cartilage healing

- Hyaline cartilage透明軟骨
  - Injury to the cartilage surface (partial thickness部分厚度)
  - Injury into subchondral bone (full thickness全厚度)
  - Finally OA (osteoarthritis骨性關節炎)
- The recovery of the cartilage depends on the duration of immobilization and its condition
  - After 3-4 months of immobilization, it takes almost 1 year for young dogs to achieve complete cartilage regeneration
- 軟骨復原與關節不活動期間長短、損傷程度及動物狀態有關
- 年輕狗經3-4個月關節不活動，需幾乎1年時間復原

©Liu, Ching-Ming 2026

Healing timeline: articular cartilage

- PARTIAL THICKNESS部分厚度缺損
- Inflammation
  - No inflammatory response—avascular
  - 因無血流供應所以無炎症反應
- Repair
  - Mitotic activity ceases 1 week after injury
  - 修補期有絲分裂1週後停止
- Remodeling/Maturation
  - Local chondrocyte activity does not adequately fill the defect at 1 year because of lack of inflammatory healing response
  - 重建期即使1年軟骨細胞亦無法適當填滿軟骨缺損

©Liu, Ching-Ming 2026

Healing timeline: articular cartilage

- FULL THICKNESS (to subchondral bone)全厚度缺損達軟骨下骨
- Inflammation
  - Immediate to 3-5 days (inflammatory response from marrow cells)
  - 炎症期達3-5天，炎症細胞來自骨髓細胞
- Repair
  - 5-7 days to 2 months
  - 修補期5-7天到2個月
- Remodeling/Maturation
  - Occurs in 2-6 months; by 6 months repair tissue changed significantly from normal hyaline cartilage. Repaired tissue is fibrous and inferior to hyaline cartilage. At 1 year, 20% is type II collagen.
- 重建期始於2-6個月，以纖維組織取代透明軟骨，達1年，有20%為2型膠原纖維
- Healing/Strength Characteristics
  - Defect repair depends on patient age, size, and location of the defect. Regardless, complete restoration of hyaline articular cartilage is very rare.
- 透明軟骨極少完全復原

©Liu, Ching-Ming 2026

Implications for Rehabilitation of articular cartilage

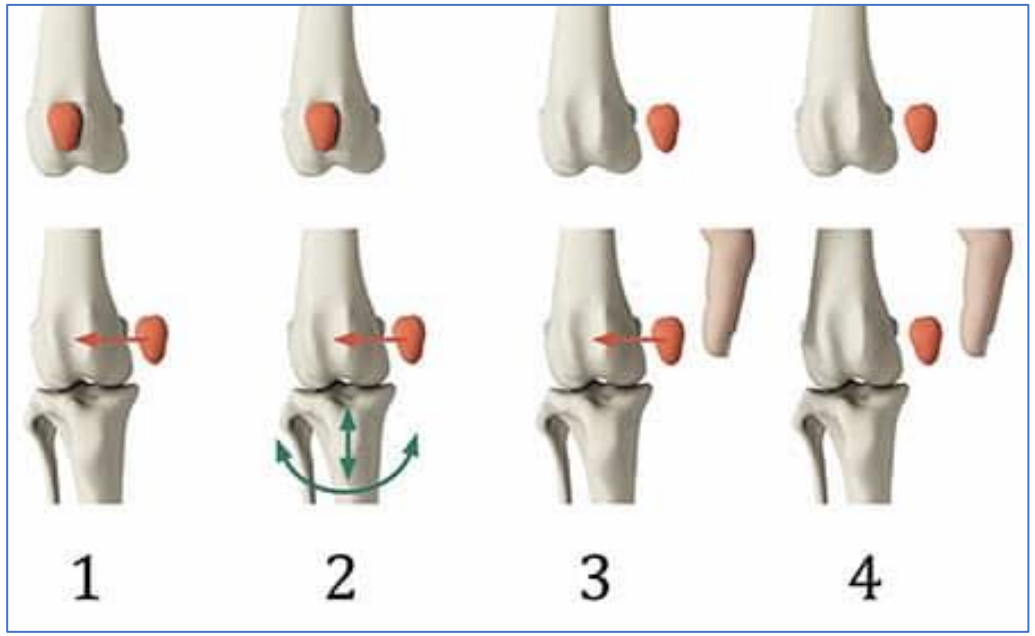
- Continuous passive range of motion (PROM) after injury appears to improve healing with mainly hyaline cartilage observed in 60% of defects (adolescent) and 44% of defects (adult) in rabbits undergoing continuous PROM, compared to 10% or fewer in controls (immobilization in a cast or free range in cages).
- Less profound results occur in defects larger than 3 mm in diameter.
- 一項研究進行持續被動式關節活動能增進透明軟骨缺損癒合，未成年兔60%，成年兔44%，反觀對照組以石膏固定限制關節活動復原僅10%或更低

©Liu, Ching-Ming 2026

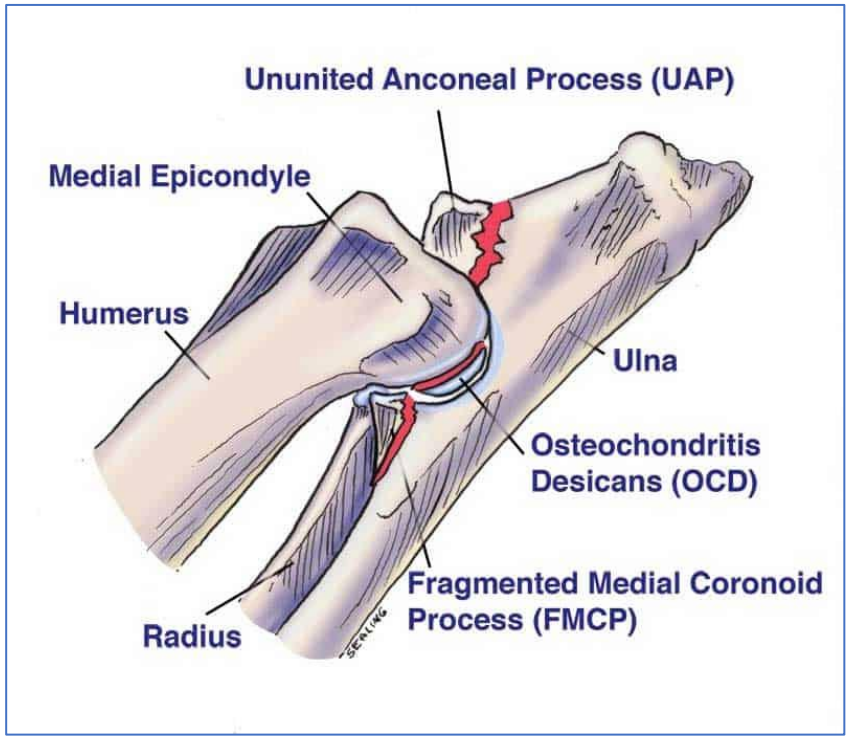
- Considerations for Remobilization of Cartilage after Immobilization
  - The length of immobilization, condition of cartilage, and the length and magnitude of weight bearing after immobilization affect cartilage recovery
  - After a 6-week immobilization period, 3 weeks of free, low-intensity activity results in normal cartilage
  - Longer periods of immobilization require longer recovery times
  - Immobilization longer than 15 weeks may **not** result in complete recovery, even with 50 weeks of remobilization, in young dogs
  - Vigorous exercise after immobilization may be deleterious to cartilage
- 固定關節使軟骨復原極受影響，固定關節不活動時間愈長所需復原時間愈長。在年輕狗，超過15週固定不活動即使之後再做50週活動，軟骨無法完全復原，解除固定後做劇烈運動對軟骨有害

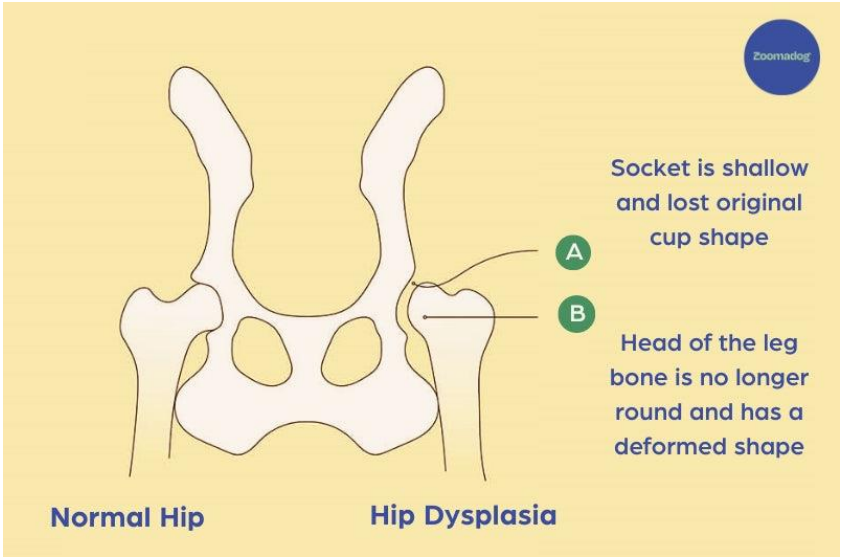
©Liu, Ching-Ming 2026

- Elbow Dysplasia
- MPL (Medial Patellar Luxation)
- HD (Hip Dysplasia)



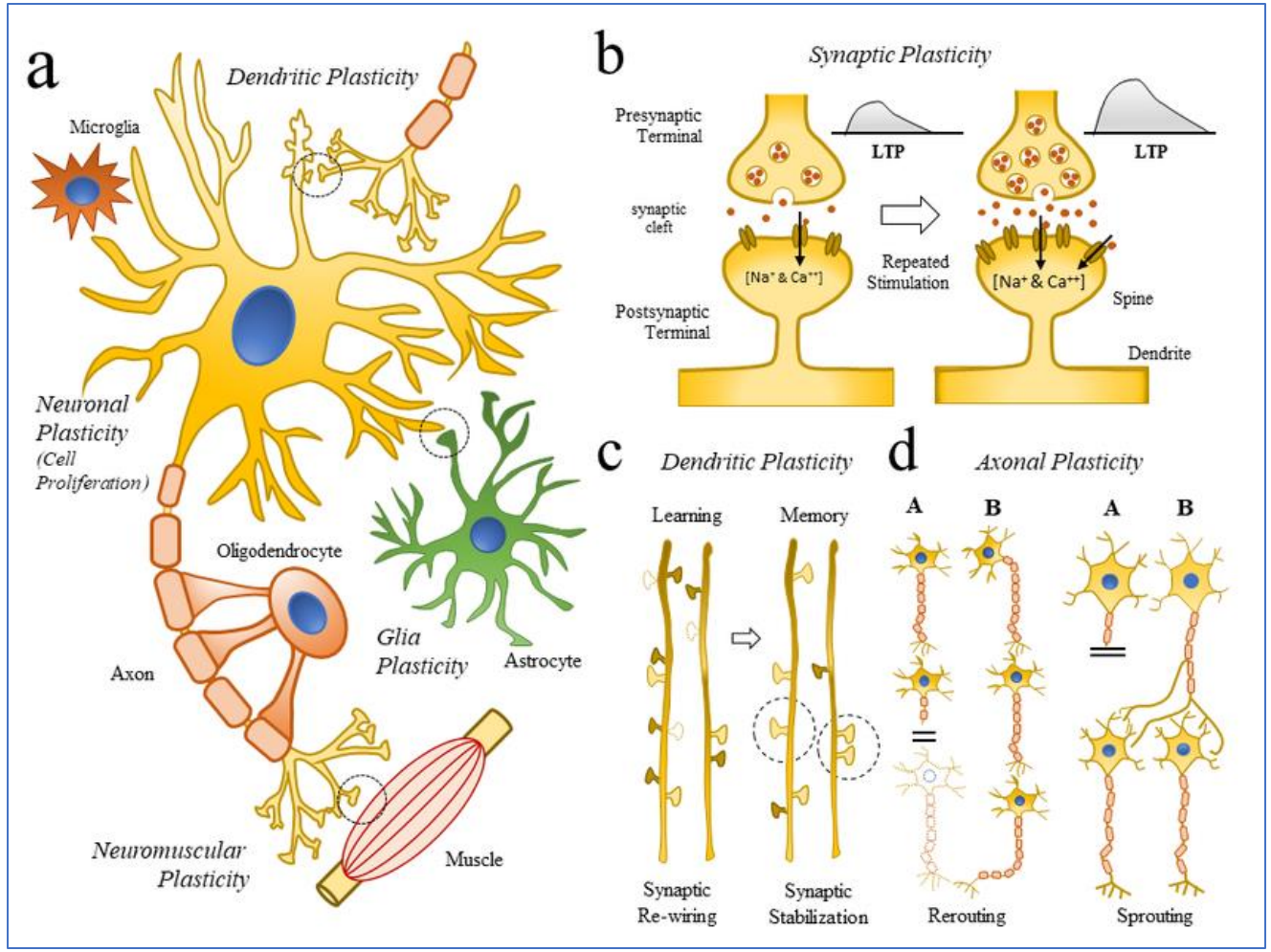
In-in in-out out-in out-out





©Liu, Ching-Ming 2026

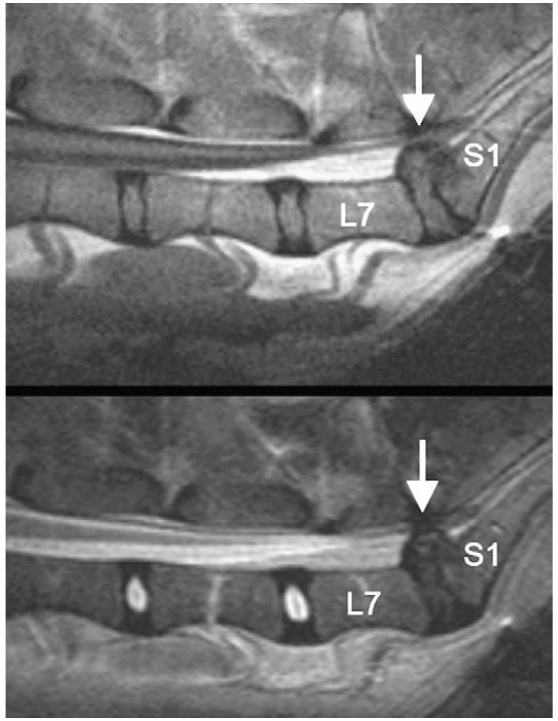
- CNS recovery is not possible, the nervous system can undergo plastic changes of the neural circuits allowing impressive regain of function.
- 中樞神經系統（CNS）在遭受不可逆的損傷後便無法恢復，但神經系統可以讓神經迴路產生塑性改變，從而實現令人驚嘆的功能恢復。主要有兩種可能的方式
  - Synaptic plasticity 突觸可塑性
    - a normal strategy of the CNS to react to changes (supersensitivity).
    - changes to the ion channels or the generation if new synapses. The generation of new synapses can be positively influenced by rehabilitation
  - Synaptogenesis 突觸形成 and collateral sprouting of intact afferents
- Animals are also capable of learning new ways of moving
  - " spinal walking" is a random, rhythmical and alternating motion of the back legs, without supraspinal or cortical control of the movement

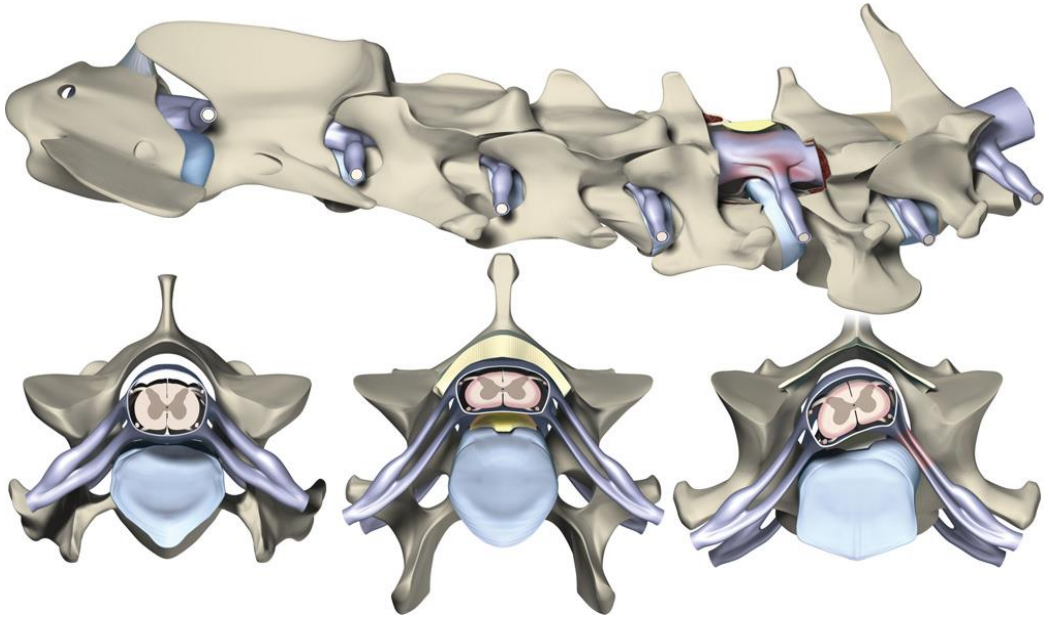


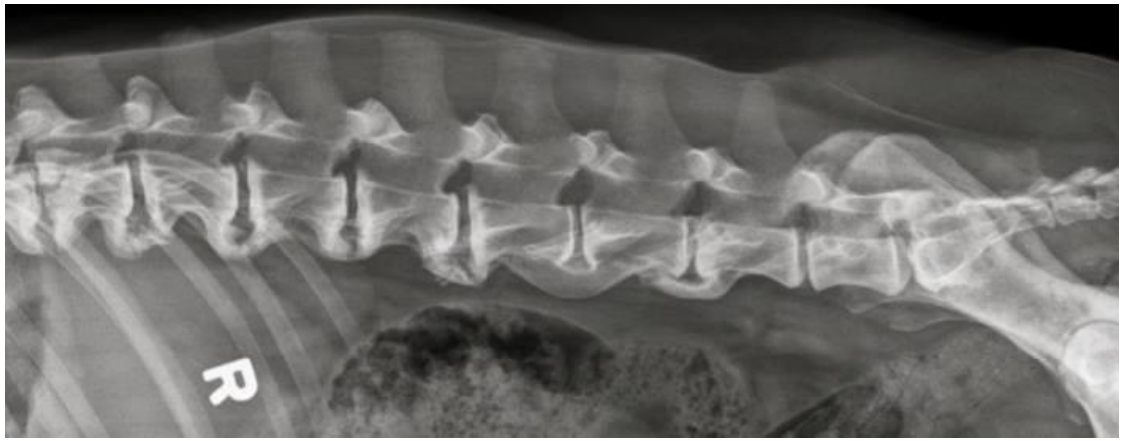
J. Integr. Neurosci. 2020 vol. 19(3), 571–592

Nerve

- Central nervous system 中樞神經系統
  - Vertebral body fractures or displacement
  - Disc herniation
  - DLSS (Degenerative Lumbosacral Stenosis)
  - Deep pain sensation is the key of prognosis





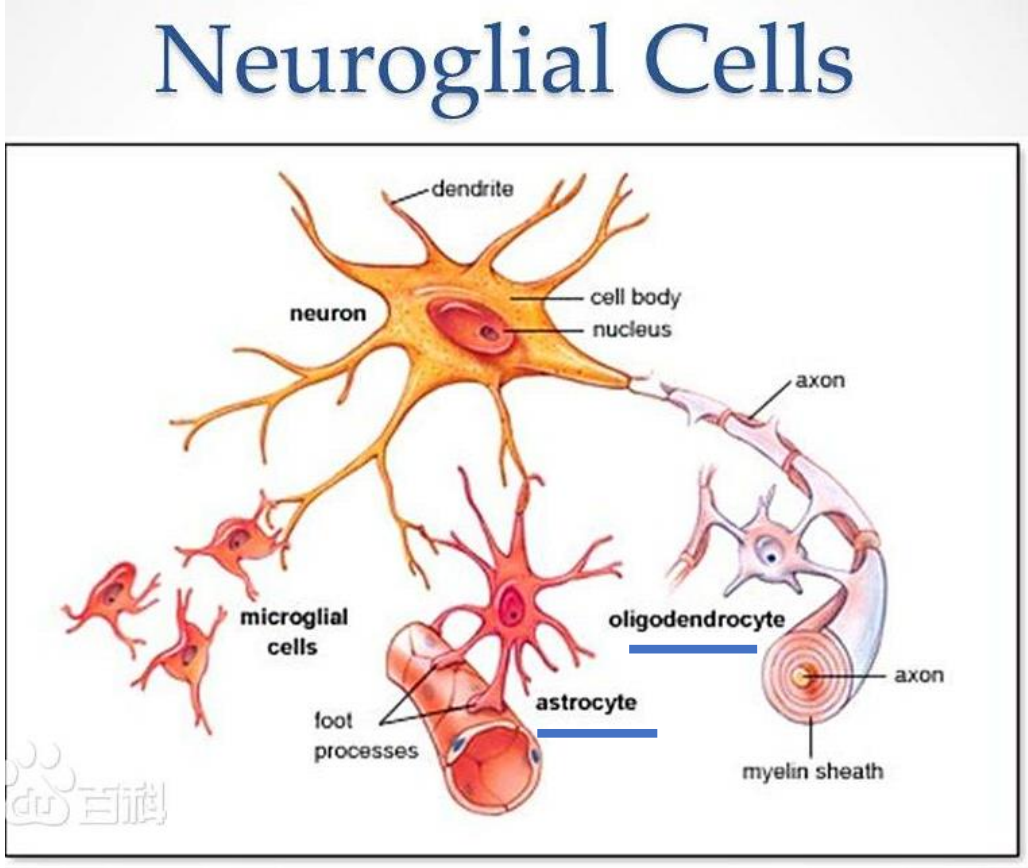


https://www.google.com/

©Liu, Ching-Ming 2026

Spinal cord trauma 脊髓創傷

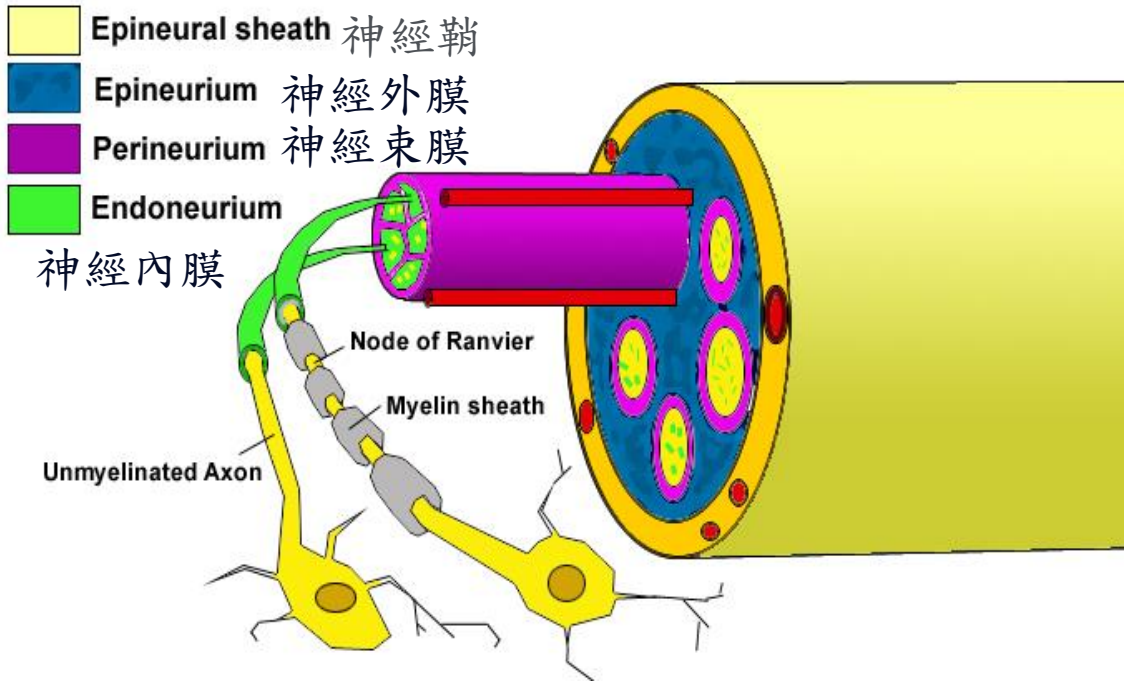
- The mechanical contusion generates secondary injuries of the neurons and the glial cells (necrosis and apoptosis).
- The initial inflammatory response has two phases of cellular infiltration:
  - neutrophils, with a peak within a few hours
  - followed by macrophages that peak after about one week.
- Secondary cellular events seem to be more important than the initial mechanical contusion as they can develop a long time after the initial injury.
- The secondary mechanisms of injury after a trauma consist of a series of both biochemical and metabolic events that may last for weeks.
- 機械式挫傷最初引起嗜中性球浸潤隨後巨噬細胞出現受傷後一周達高峰。
- 繼發性的細胞事件似乎比最初的機械性挫傷更為重要，因為這些變化可能在初始損傷發生很久之後才逐漸出現。創傷後的繼發性損傷機制包括一連串的生化與代謝變化，而這些變化可能持續數週。
- Oligodendrocytes (寡突膠質細胞) and astrocytes (星形膠質細胞) can be regenerated in the CNS.
  - The destruction of oligodendrocytes leads to a destruction of the myelin sheath of the axons resulting in a loss of the conduction properties. There has been some evidence showing that slowed function can be retrained even without the myelin sheath of the axons.
  - Astrocytic proliferation and hypertrophy follows a series of events like axon degeneration, inflammation and trauma. This astrocytosis (gliosis) can generate a scar in the CNS, acting as a barrier for the regrowth of the axons.



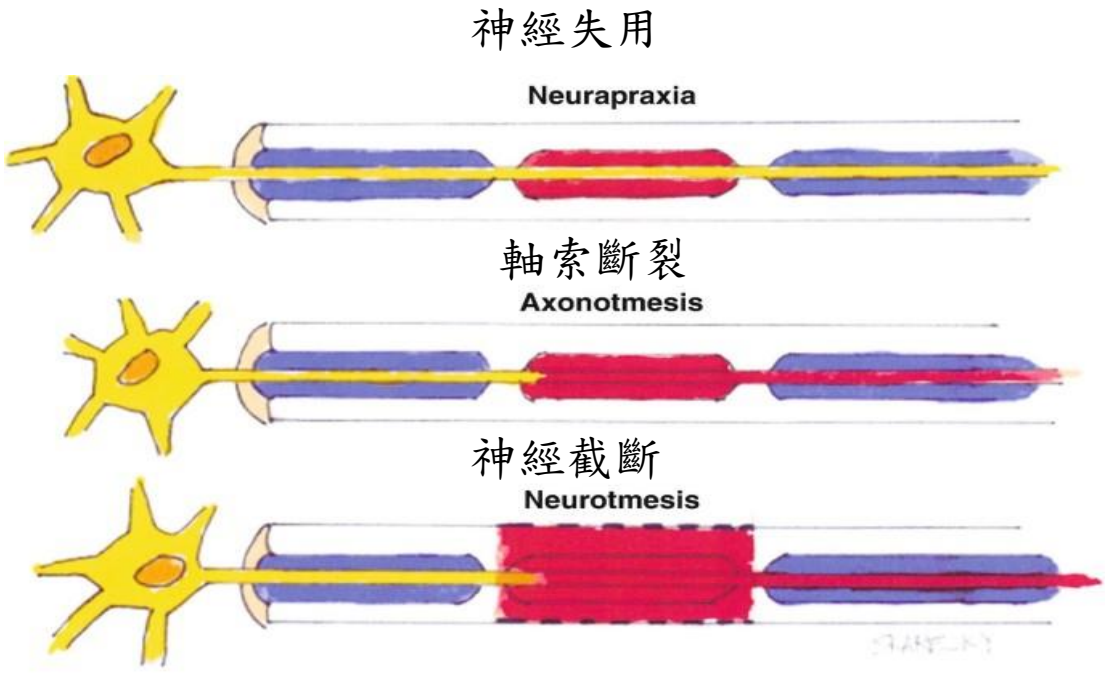
https://www.google.com/

Nerve

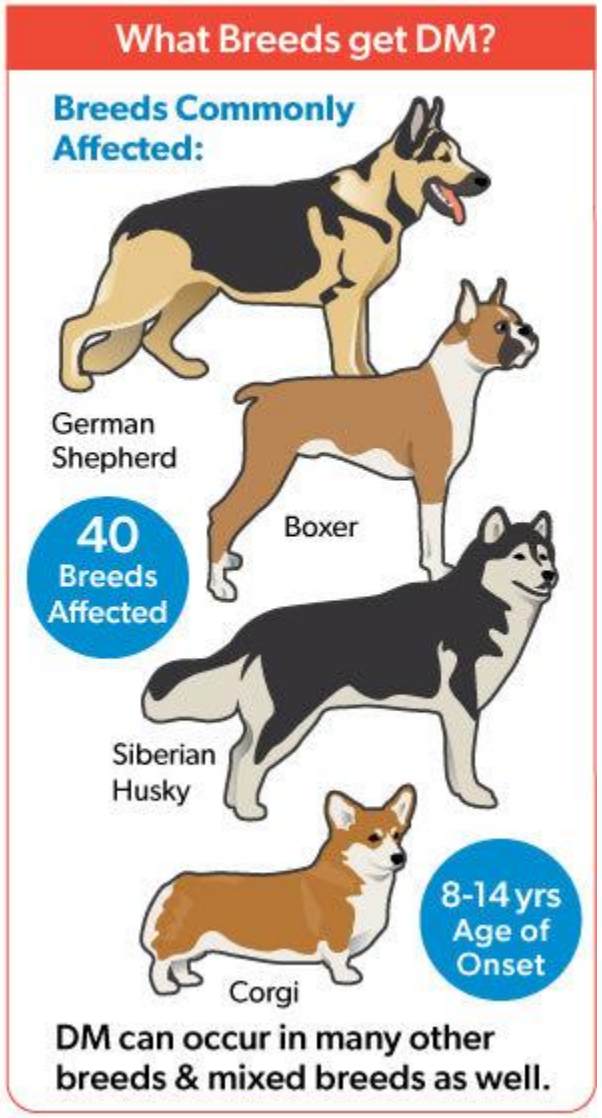
- Peripheral nervous system周邊神經系統



https://www.orthobullets.com/hand/6066/peripheral-nerve-injury-and-repair



- Degenerative myelopathy



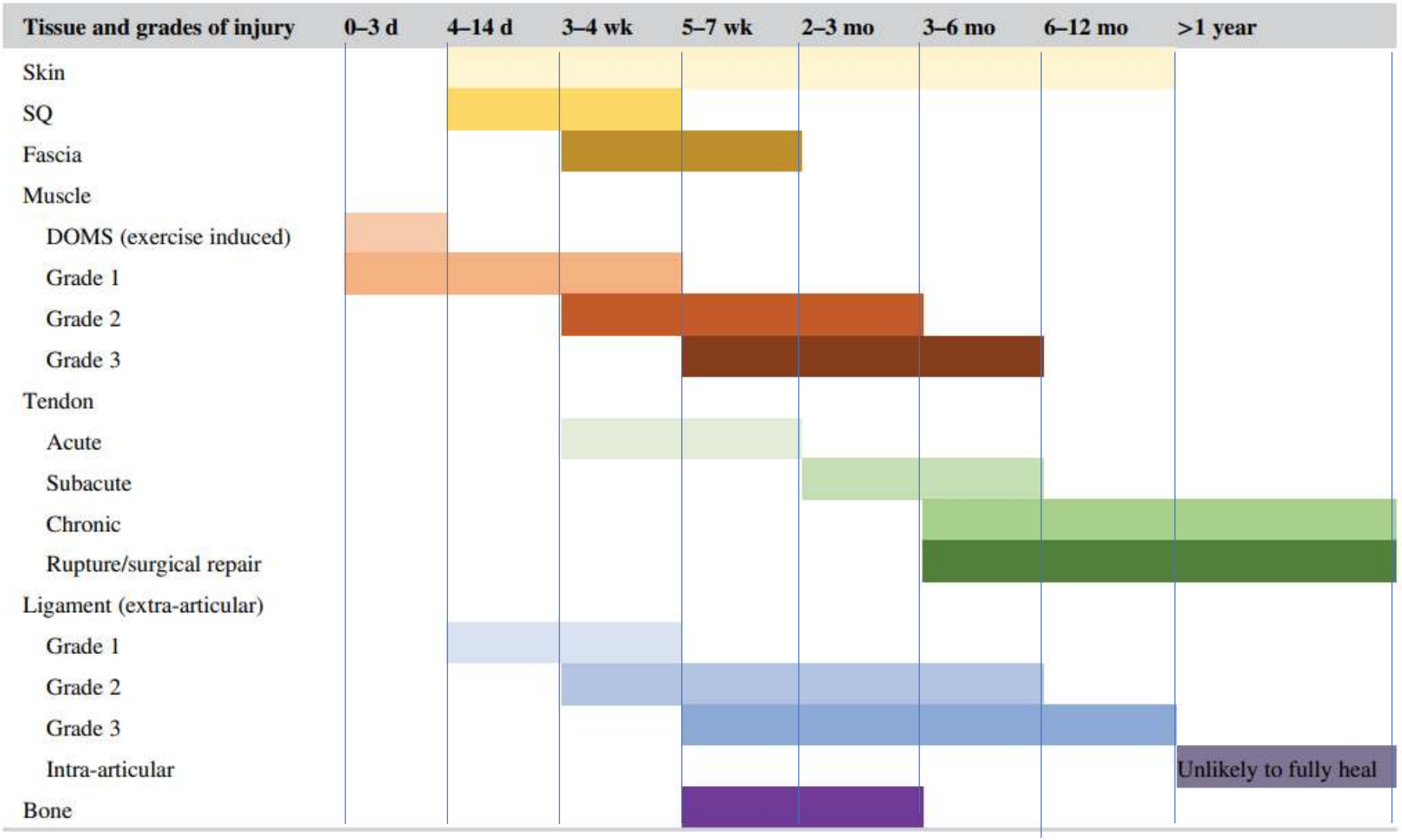
- Brachial plexus avulsion



©Liu, Ching-Ming 2026

To review fundamental principles of tissue healing and physical rehabilitation as they apply to dogs **recovering from cranial cruciate ligament (CCL) surgery**

TABLE 2 Approximate rates of tissue healing[a]



幾乎無法癒合

Fundamental principles of rehabilitation and musculoskeletal tissue healing, Veterinary Surgery 2020, K. Kirby Shaw

©Liu, Ching-Ming 2026